HKUSPACE
香港大學專業進修學院
HKU School of Professional and Continuing Education

Business Process Automation with VBA and Python

Mr. Eddie Chow / 14 Nov 2026

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- What is Process Automation, RPA, IPA?
- Overview of the technological building blocks related to process automation
- Contemporary tools for process automation
- Challenges and opportunities of business process automation
- Business implications of process automation

Intended Learning Outcomes

1. describe the contemporary trends of process automation and explain the opportunities and challenges of business process automation;
2. outline key steps in process automation project management and illustrate the significance of each step;
3. apply computational tools to implement process automation;
4. discuss the development of process automation and practical cases for business.

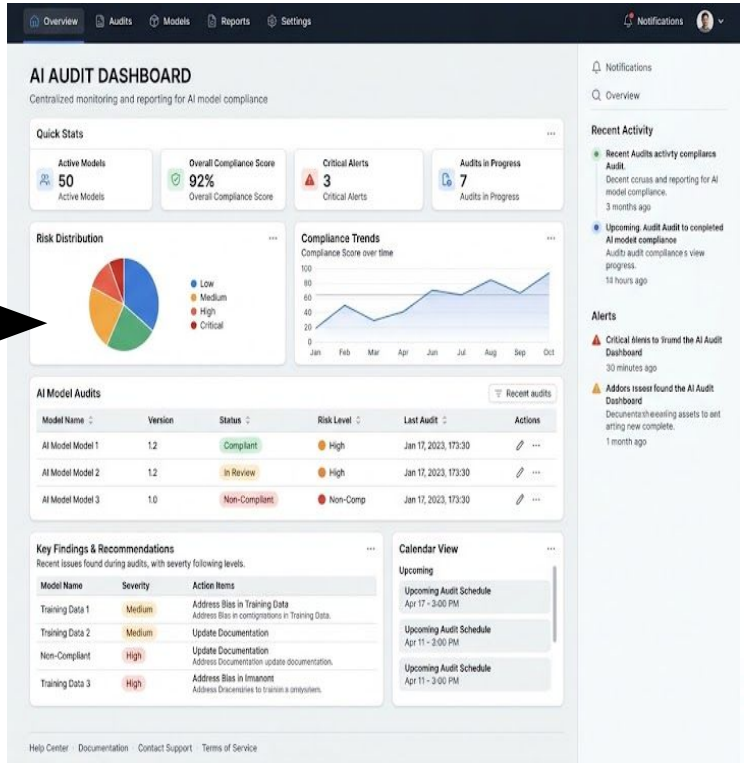
Agenda

1. Data Driven Business Strategy
2. Key Performance Indicators (KPIs)
3. Data Visualization
4. Case Studies of Successful Automation Projects
5. Unstructured Data Processing
6. Private Data Collection - Reading Data from Gmail, Google Sheet
7. Introduction to Project Management in Business Process Automation
8. Methodologies and Key Steps in Project Management
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10. Organizational Issues in Business Process Automation
11. Projects 7 Project Scope and Requirements Gathering
12. Risk Management in Automation Projects
13. Change Management Best Practices
14. Agentic AI Workflow Automation
15. Conclusion and Future Directions

Data-Driven Business Strategy

A **data-driven business strategy** is a systematic plan that guides business decisions, operational improvements, and growth using verified data analysis rather than habit or intuition.





Data-Driven vs Scenario-Driven Decision Making

Choosing the right approach for planning, execution, and risk management.

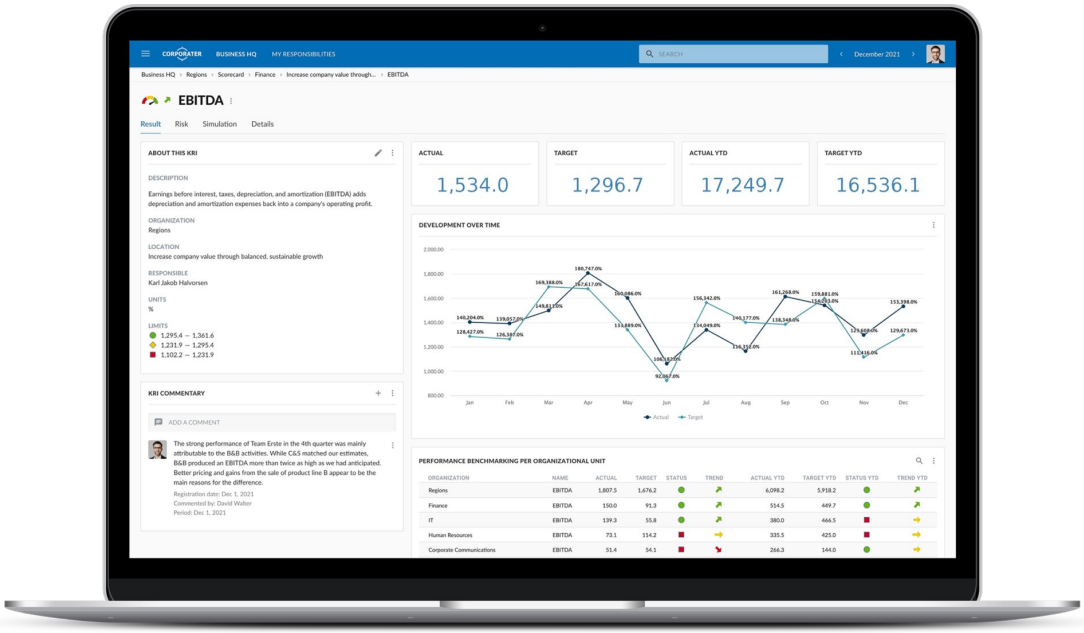
Aspect	Data-driven	Scenario-driven
Starting point	Observed data (logs, metrics, events)	Hypothetical or expected situations / “stories”
Main driver	Statistical patterns and KPIs	Business or operational logic of a situation
Best for	Optimization, performance tracking, and evidence-based management.	Strategic planning, experimentation, and uncertain environments.
Typical questions	“What is happening now?” What does the data predict?”	“What could happen?” “What should we do if it does?”

Use Case - Investment portfolio monitoring

Portfolio monitoring is used by VC, PE, hedge fund, wealth, and asset-management teams to follow performance, health, and risk after investment, while AI/ML helps analyze large financial datasets, detect patterns, and improve risk prediction.

Typical use case

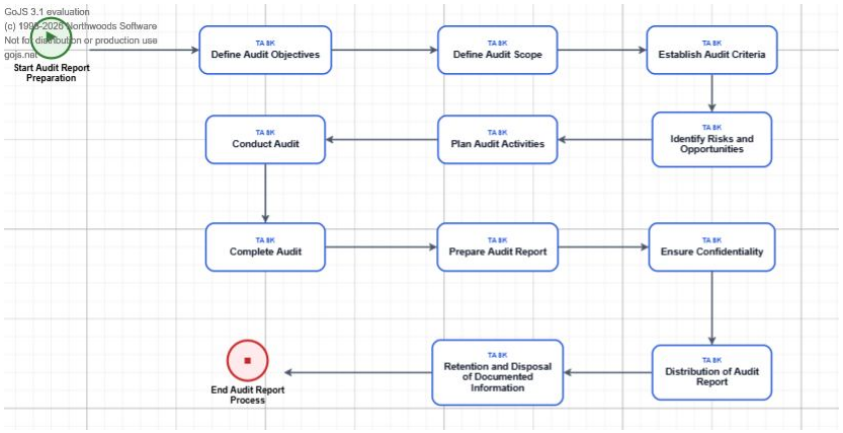
A portfolio monitoring system ingests holdings, prices, cash flows, sector exposure, and company KPIs, then checks them against policy targets and benchmarks. It can automatically flag issues like concentration risk, style drift, liquidity problems, covenant breaches, or underperforming positions.

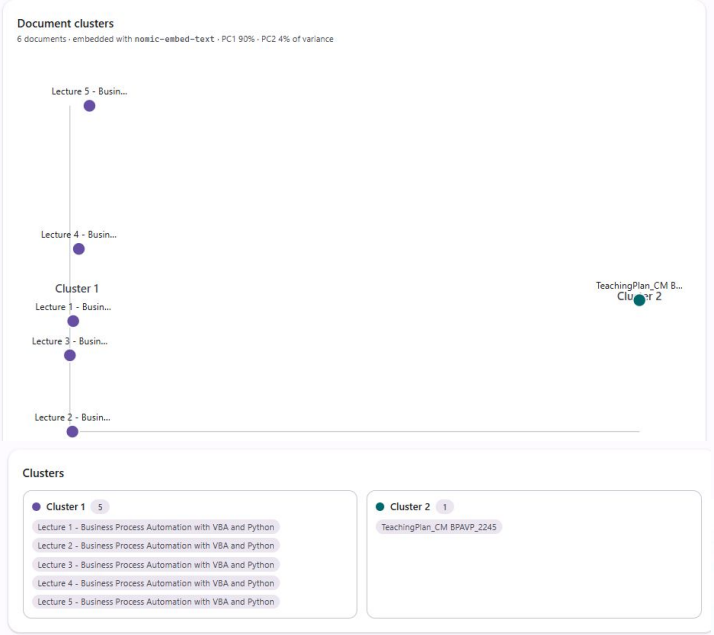


Use Case - Investment portfolio monitoring

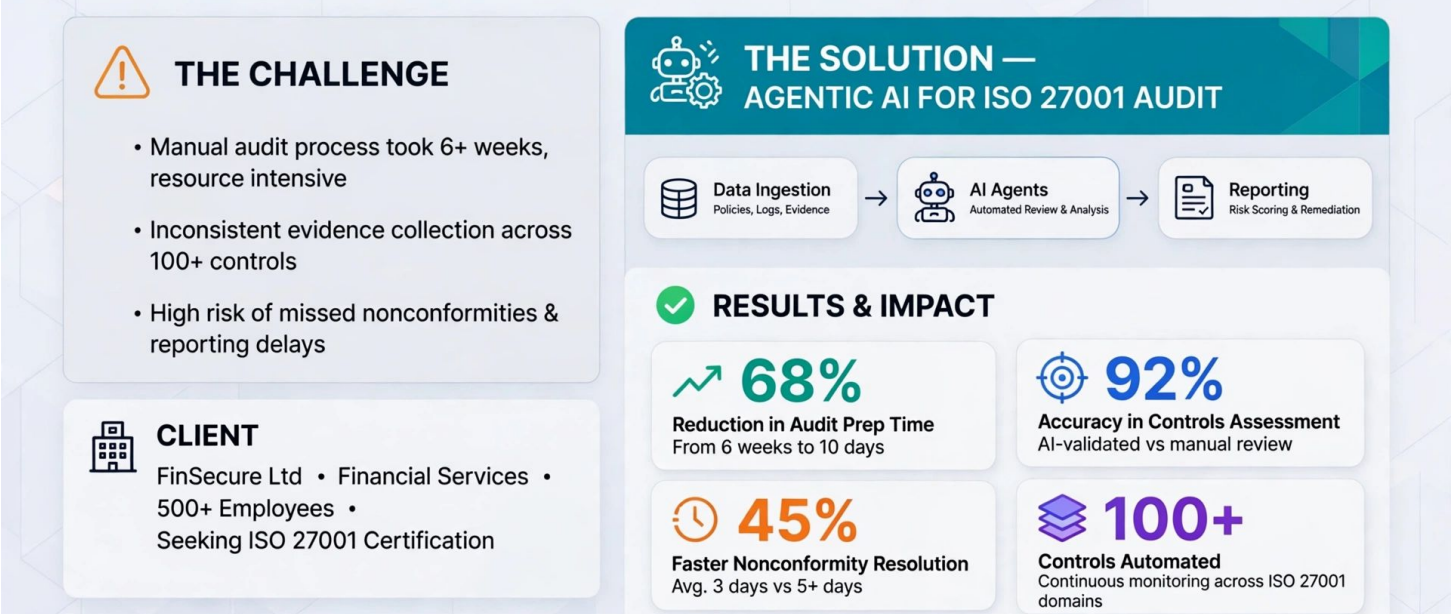
How ML and algorithms work together

- Algorithms handle deterministic logic: threshold alerts, rebalancing rules, benchmark comparisons, and risk-limit checks.
- Machine learning handles pattern detection: anomaly detection, predictive risk signals, and scoring which holdings are likely to deteriorate or outperform.
- The output is usually an alert, dashboard, or action recommendation for an investor, analyst, or portfolio manager.



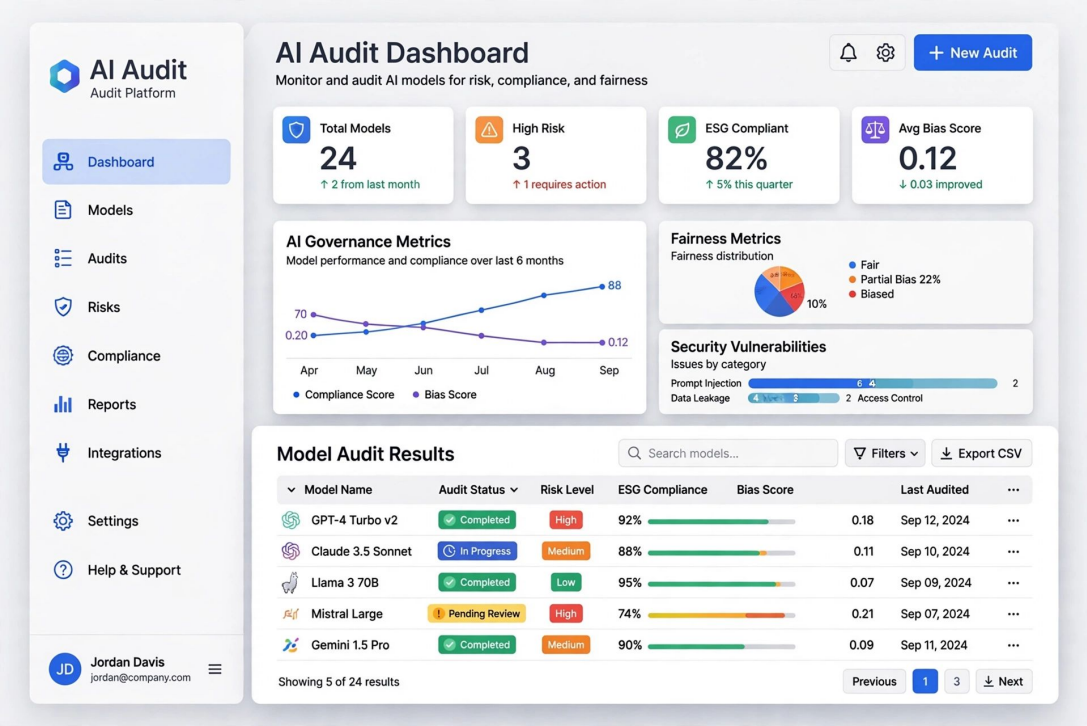


Use Case - ISO Audit with Agentic AI



Finding ID	Classification	Risk Category (Q/S/E)	Process Step	Activity	Hazard/Failure Mode	S (1-5)	L (1-5)	RPS (SxL)	Risk Level (per Matrix)
BP-01	Best Practice (Conformity)	Q	3.4	Continuous Thermal Envelope Surveillance	Chiller drift / loss of control (prevented by design)	5	1	5	Low
BP-02	Best Practice (Conformity)	Q	3.1-3.6	End-to-end process	Incomplete evidence / traceability	4	1	4	Low
OFI-01	Opportunity for Improvement	Q	3.4-3.5	Long dwell monitoring	Slow drift / delayed trend detection	4	2	8	Medium
OFI-02	Opportunity for Improvement	S	3.2	Transfer operations	Cold stress during door-open peaks	3	2	6	Medium
NC-IMPLIED-01	No NC Identified (Sampled)	Q/S/E	All	Audited scope	N/A	1	1	1	Low

Enterprise Light Mode (ESG Compliance, Bias, Risk)

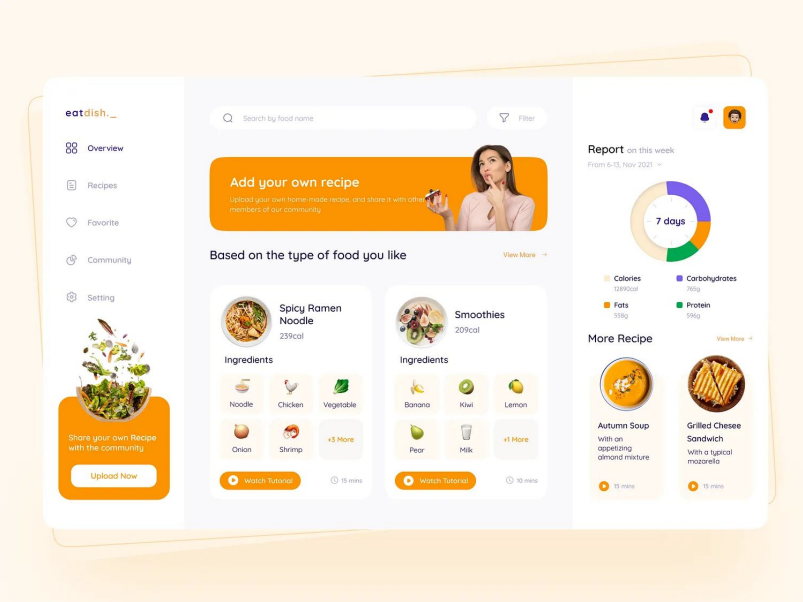


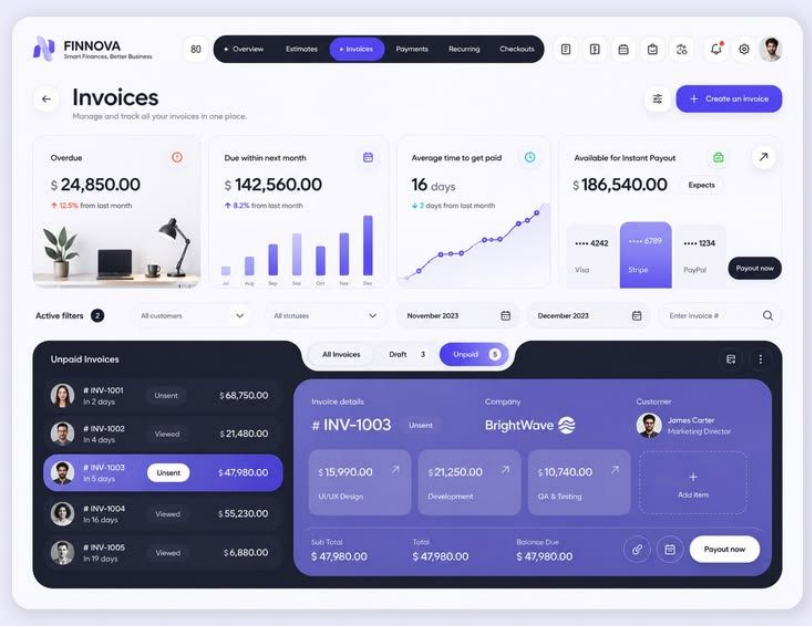


Dashboard

https://dribbble.com/search/food-dashboard

https://www.pinterest.com/ideas/kpi-dashboard/915256809649/





Safety ROI & Key Performance Indicators

- Cost Reduction: 28% via automation
- Schedule Adherence: +14% improvement
- Safety Reporting Accuracy: 99.4%
- Labor Efficiency: +22% Year-over-Year



Gemini to generate chart design

Procedure:

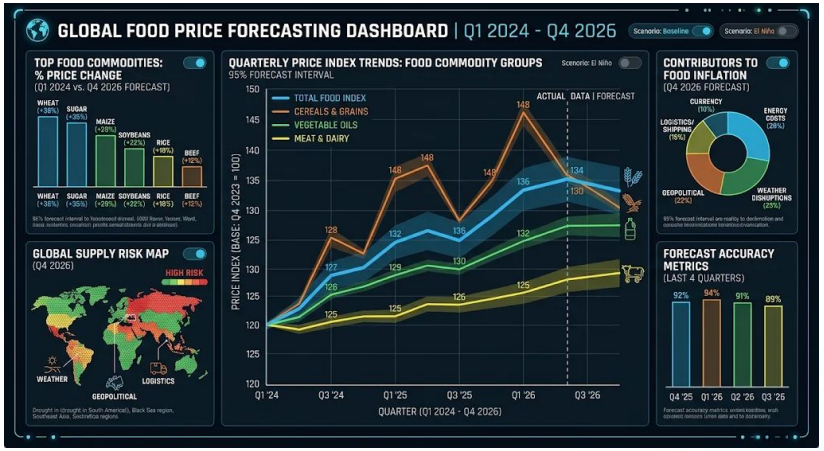
1. Open Gemini.google.com with gmail login
2. enter prompt

Create dashboard chart images with food forecasting price include title, x-axis name, y-axis name.



3. Enter another prompt with more details

Create dashboard chart images with food forecasting price with period Q1 2024 - Q4 2026 include title, x-axis name, y-axis name.



Gemini to generate python plotly code from dashboard design

Procedure:

1. Search suitable chart for visualisation from dribble, pinterest.
2. Open Gemini.google.com with gmail login
3. Upload the chart images and enter prompt

Based on above images, generate python plotly code include title, x-axis name, y-axis name.



Python

```
import plotly.graph_objects as go

# Data approximated from the dashboard chart for February 2022
dates = ['Feb 1, 2022', 'Feb 8, 2022', 'Feb 15, 2022', 'Feb 22, 2022', 'Feb 28, 2022
revenue = [3200, 2800, 4251, 3500, 4100]

fig = go.Figure()

fig.add_trace(go.Scatter(
    x=dates,
    y=revenue,
    mode='lines+markers',
    name='Revenue',
    line=dict(color='#10b981', width=3),
    marker=dict(size=8, color='#10b981')
))

fig.update_layout(
    title='Total Revenue',
    xaxis_title='Date',
    yaxis_title='Revenue ($)',
    template='plotly_white',
    hovermode='x unified'
)

fig.show()
```

Visualization using Programming

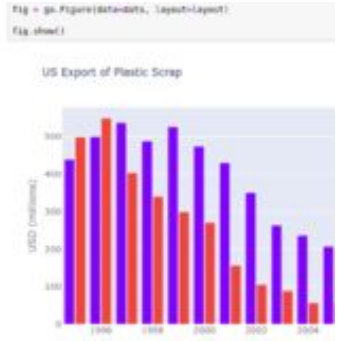
- Python
  - **matplotlib**
  - seaborn
  - **plotly(mainly)**
  - pylab
- R
  - graphics
  - ggplot2

Plotly: Visualization with Python

Plotly's Python graphing library makes interactive, publication-quality graphs.

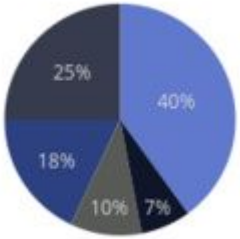
Examples of how to make line plots, scatter plots, area charts, bar charts, error bars, box plots, histograms, heatmaps, subplots, multiple-axes, polar charts, and bubble charts.
Plotly.py is free and open source..

Basic Chart: https://plotly.com/python/basic-charts/



Features:

1. Create publication quality plots.
2. Make interactive figures that can zoom, pan, update.
3. Customize visual style and layout.
4. Export to many file formats.
5. Embed in JupyterLab and Graphical User Interfaces.
6. Use a rich array of third-party packages built on Matplotlib.



https://plotly.com/python/

Plotly: Visualization with Python

The main modules in Plotly are:

- plotly.plotly
- **plotly.graph.objects**
- plotly.express

- plotly.plotly acts as the interface between the local machine and Plotly. It contains functions that require a response from Plotly's server.

- **plotly.graph_objects module contains the objects (Figure, layout, data, and the definition of the plots like scatter plot, line chart) that are responsible for creating the plots.**

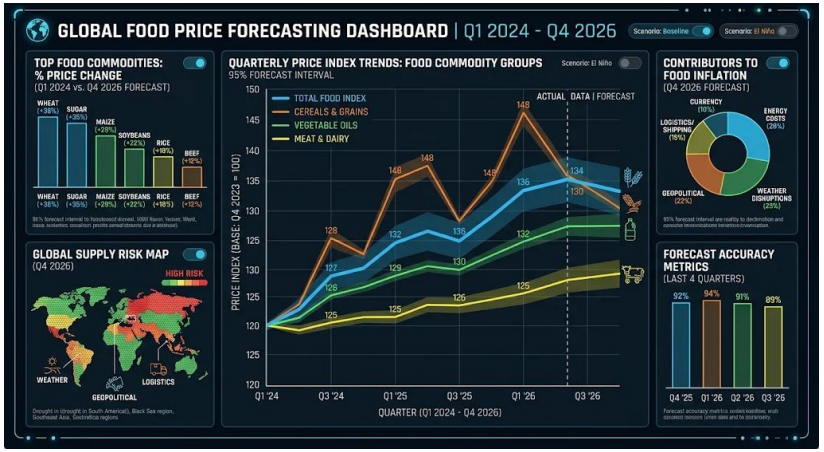
- plotly.express module can create the entire Figure at once. It uses the graph_objects internally and returns the graph_objects.Figure instance.

Gemini to generate python plotly code from dashboard design

Procedure:

1. Search suitable chart for visualisation from dribble, pinterest.
2. Open Gemini.google.com with gmail login
3. Upload the chart images and enter prompt

Based on above images, generate python plotly plotly.graph.objects code include title, x-axis name, y-axis name.




```python
import plotly.graph_objects as go

# Define quarters for the period Q1 2024 - Q4 2026
quarters = [
    "Q1 '24", "Q2 '24", "Q3 '24", "Q4 '24",
    "Q1 '25", "Q2 '25", "Q3 '25", "Q4 '25",
    "Q1 '26", "Q2 '26", "Q3 '26", "Q4 '26"
]

# Data extracted from the Global Food Price Forecasting Dashboard image
total_food_index = [120, 122, 125, 127, 129, 131, 132, 130, 136, 148, 136, 134]
cereals_grains = [120, 123, 125, 128, 132, 135, 140, 130, 148, 148, 130, 128]
vegetable_oils = [120, 121, 124, 126, 129, 131, 130, 129, 132, 133, 132, 131]
meat_dairy = [120, 121, 122, 125, 125, 126, 127, 126, 125, 127, 128, 129]

# Create figure using plotly.graph_objects
fig = go.Figure()

# Add traces corresponding to the dashboard chart lines
fig.add_trace(go.Scatter(
    x=quarters, y=total_food_index,
    mode='lines+markers',
    name='TOTAL FOOD INDEX',
    line=dict(color='#00e5ff', width=3)
))

fig.add_trace(go.Scatter(
    x=quarters, y=cereals_grains,
    mode='lines+markers',
    name='CEREALS & GRAINS',
    line=dict(color='#ff7f0e', width=3)
))
```

Plotly Exercise 1

```
#using plotly, display histgram chart and download chart
import numpy as np
import plotly.graph_objects as go

# Generate sample data for the sine wave
x = np.linspace(0, 10, 100)  # 100 points from 0 to 10
y = np.sin(x)                # y is the sine of x

# Generate sample data for the histogram
data = np.random.normal(loc=0, scale=1, size=1000) #
    random values from a normal distribution

# Create a plotly figure
fig = go.Figure()

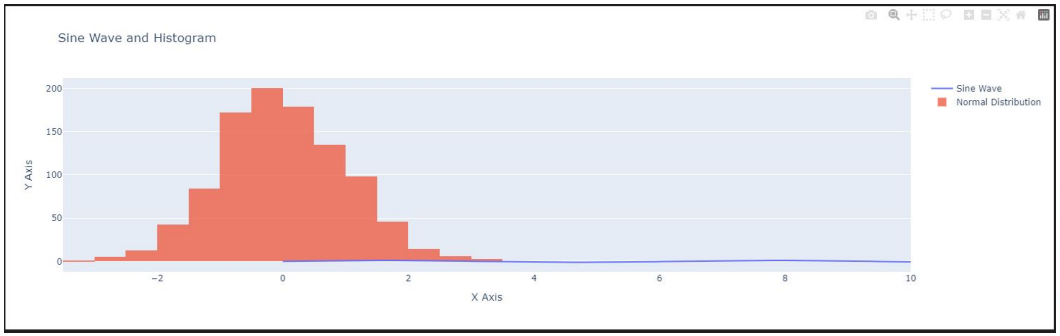
# Add a line plot for the sine wave
fig.add_trace(go.Scatter(x=x, y=y, mode='lines',
    name='Sine Wave'))
```

```
# Add a histogram
fig.add_trace(go.Histogram(x=data, name='Normal
Distribution', opacity=0.75, nbinsx=30))

# Customize layout
fig.update_layout(title='Sine Wave and Histogram',
                  xaxis_title='X Axis',
                  yaxis_title='Y Axis',
                  barmode='overlay')  # Overlay histograms

# Display the interactive plot in your browser or notebook
fig.show() # Save/download the figure as a PNG file

# (Note: Requires kaleido installed via: pip install -U
kaleido) fig.write_image('sine_wave_histogram.png')
print("Chart saved successfully as
'sine_wave_histogram.png'")
```



Plotly Example - Line Chart

```
import plotly.graph_objects as go

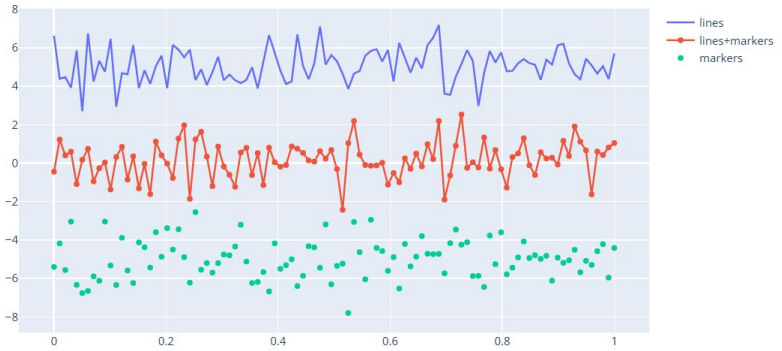
# Create random data with numpy
import numpy as np
np.random.seed(1)

N = 100
random_x = np.linspace(0, 1, N)
random_y0 = np.random.randn(N) + 5
random_y1 = np.random.randn(N)
random_y2 = np.random.randn(N) - 5

# Create traces
fig = go.Figure()
fig.add_trace(go.Scatter(x=random_x, y=random_y0,
                         mode='lines',
                         name='lines'))
fig.add_trace(go.Scatter(x=random_x, y=random_y1,
                         mode='lines+markers',
                         name='lines+markers'))
fig.add_trace(go.Scatter(x=random_x, y=random_y2,
                         mode='markers', name='markers'))

fig.show()
```

Can be find in
https://plotly.com/python/line-charts/



Plotly Example - Bar Chart

```
import plotly.graph_objects as go

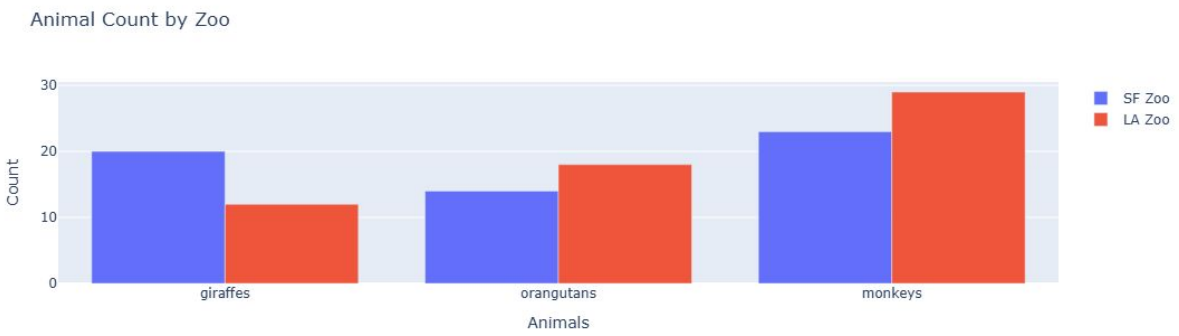
animals = ['giraffes', 'orangutans', 'monkeys']

fig = go.Figure(data=[
    go.Bar(name='SF Zoo', x=animals, y=[20, 14, 23]),
    go.Bar(name='LA Zoo', x=animals, y=[12, 18, 29])
])

# Change the bar mode
fig.update_layout(
    barmode='group',
    title='Animal Count by Zoo',
    xaxis_title='Animals',
    yaxis_title='Count'
)

fig.show()
```

Can be find in https://plotly.com/python/bar-charts/



Plotly Example - Scatter Chart

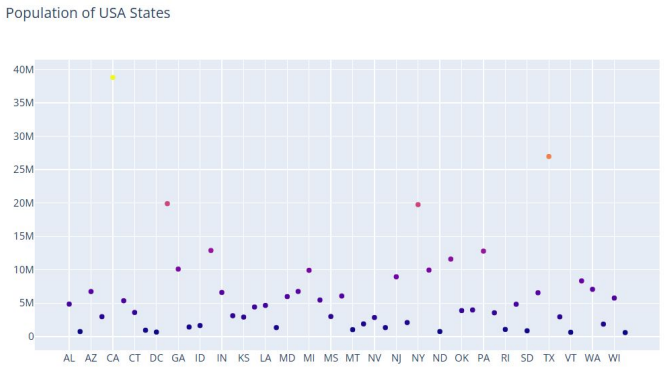
```
import plotly.graph_objects as go
import pandas as pd

data= pd.read_csv("https://raw.githubusercontent.com/plotly/datasets/master/2014_usa_states.csv")

fig = go.Figure(data=go.Scatter(x=data['Postal'],
                                y=data['Population'],
                                mode='markers',
                                marker_color=data['Population'],
                                text=data['State'])) # hover text goes here

fig.update_layout(title=dict(text='Population of USA States'))
fig.show()
```

Can be find in https://plotly.com/python/line-and-scatter/



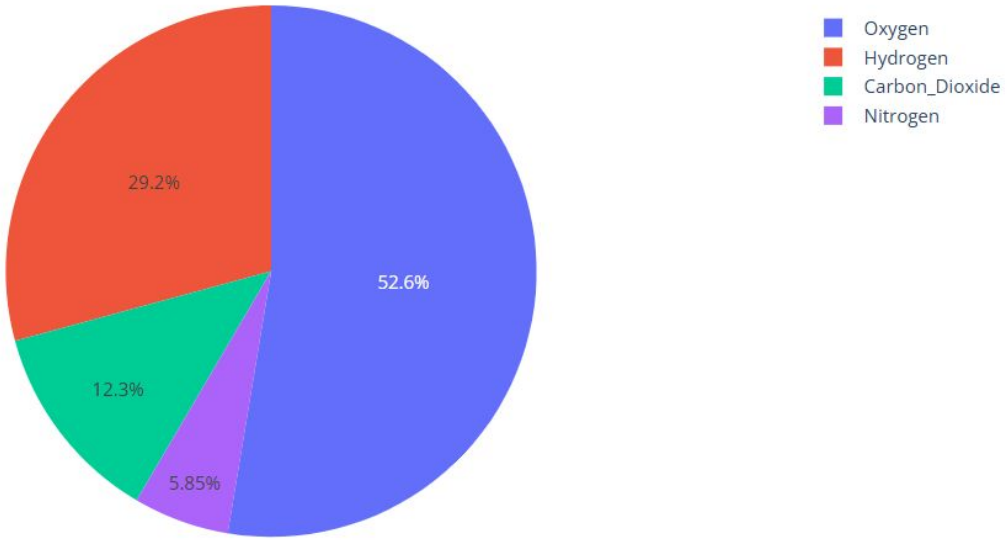
Plotly Example - Pie Chart

Can be find in https://plotly.com/python/pie-charts/

```
import plotly.graph_objects as go

labels = ['Oxygen','Hydrogen','Carbon_Dioxide','Nitrogen']
values = [4500, 2500, 1053, 500]

fig = go.Figure(data=[go.Pie(labels=labels, values=values)])
fig.show()
```



Data Type Example

	Data Type	Example	Mutable?	Ordered?	Use Case	Example Code
1	**list** list	[1, 2, 3, 2]	✓ Yes	✓ Yes	General ordered collection; allows duplicates	my_list = [1, 2, 3]
2	**tuple** tuple	(1, 2, 3)	✗ No	✓ Yes	Immutable ordered collection; hashable	my_tuple = (1, 2, 3)
3	**set** set	{1, 2, 3}	✓ Yes	No No	Unique items; fast membership tests	my_set = {1, 2, 3}
4	**dict** dict	~."a": 1, "b": 2}	✓ Yes	Yes (3.7+)	Key-value pairs; fast lookups	my_dict = {"key": "value")
5	**pandas.Series** Series	Index → Value	✓ Yes	✓ Yes	1D labeled array; like a column	pd.Series([10,20], index{"a","b")
6	**pandas.DataFrame** pandas	Table of rows/columns	✓ Yes	✓ Yes	2D tabular data; spreedsheet-like	pd.DataFrame("col": [1,2]))
6	**numpy.nndraff** numpy	array([1, 2, 3])	✓ Yes	✓ Yes	Numerical array; vectorized ops, fast math	np.array([1,2,3])
8	**json** json	~."name": "Alice", "age": 25}	No* (text)	No* (dict order)	Data interchange format; string/text	json.loads("[k": 1")

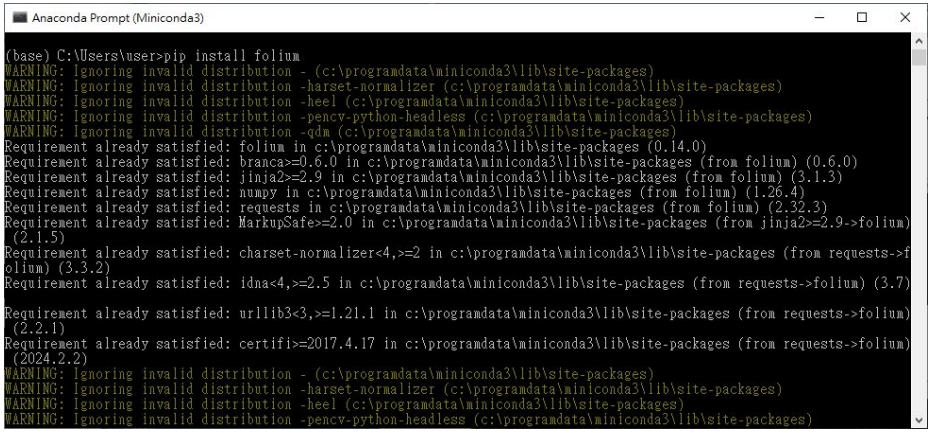
Tip: Use *list/set/dict* for general Python. Use NumPy for numerical work. Use pandas for tabular/data analysis. JSON is for storage/transport, not a native Python type.

Python Google Map Tutorial

1. Open tools “Anaconda Prompt” which will direct you to the command prompt.



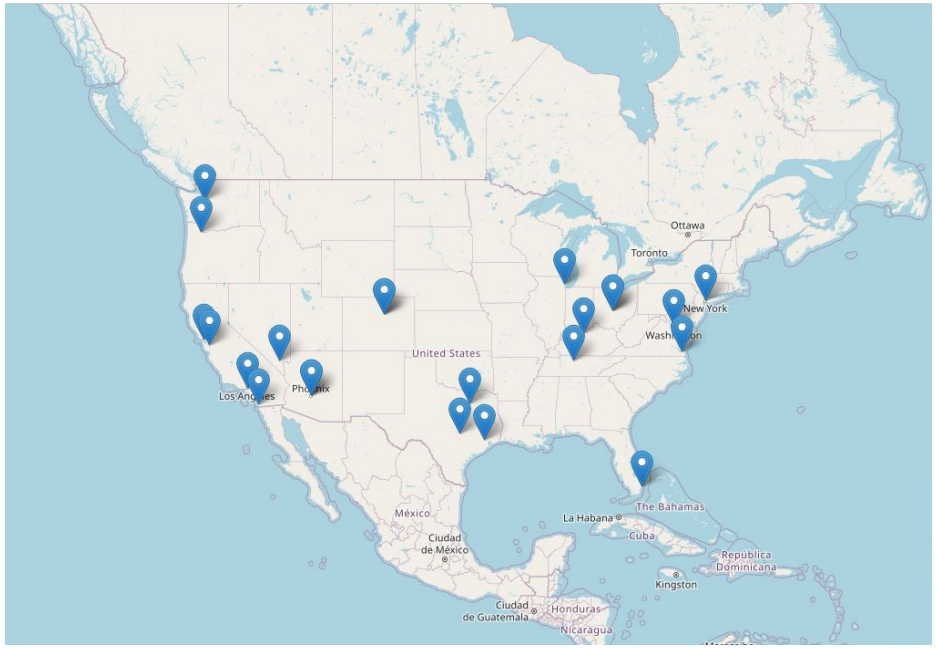
2. Install python package requests by entering “pip install folium”



Python Google Map Tutorial

```
import folium

# List of 23 coordinates (latitude, longitude)
coordinates = [
    (37.7749, -122.4194),  # San Francisco, CA
    (34.0522, -118.2437),  # Los Angeles, CA
    (40.7128, -74.0060),   # New York, NY
    (41.8781, -87.6298),    # Chicago, IL
    (29.7604, -95.3698),    # Houston, TX
    (33.4484, -112.0740),   # Phoenix, AZ
    (39.7392, -104.9903),   # Denver, CO
    (47.6062, -122.3321),   # Seattle, WA
    (38.9072, -77.0369),    # Washington, D.C.
    (39.9612, -82.9988),    # Columbus, OH
    (30.2672, -97.7431),    # Austin, TX
    (32.7767, -96.7970),    # Dallas, TX
    (37.3382, -121.8863),   # San Jose, CA
    (39.7392, -104.9903),   # Denver, CO
    (36.1699, -115.1398),    # Las Vegas, NV
    (25.7617, -80.1918),    # Miami, FL
    (38.2527, -85.7585),    # Louisville, KY
    (39.9612, -82.9988),    # Columbus, OH
    (36.8508, -76.2859),    # Norfolk, VA
    (33.4484, -112.0740),   # Phoenix, AZ
    (45.5155, -122.6793),   # Portland, OR
    (36.1627, -86.7816),    # Nashville, TN
    (32.7157, -117.1611),   # San Diego, CA
]
```



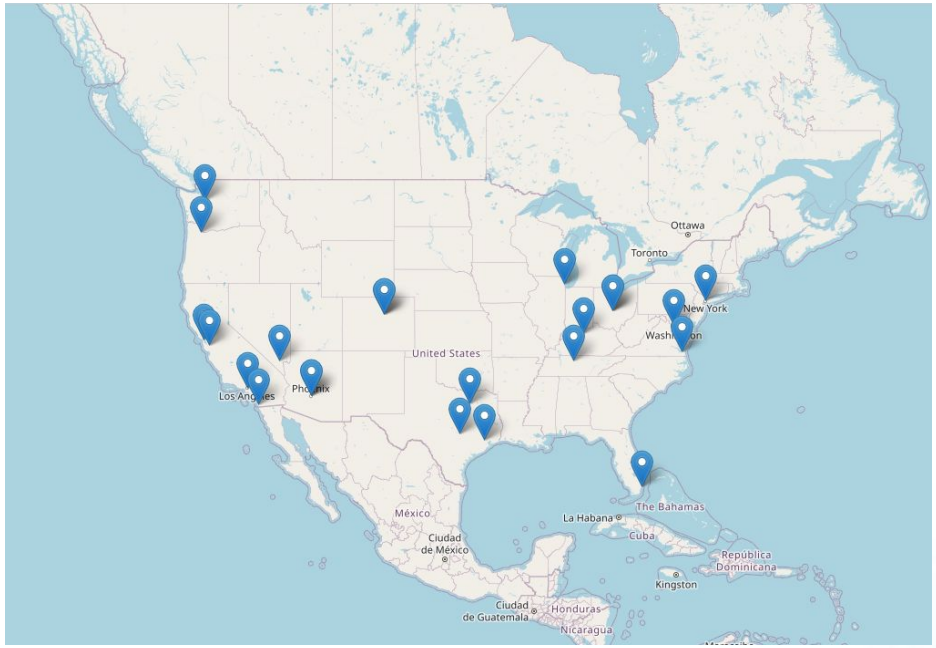
Python Google Map Tutorial

```
# Create a map centered at the first coordinate
my_map = folium.Map(location=coordinates[0],
    zoom_start=4)

# Add markers for each coordinate
for lat, lon in coordinates:
    folium.Marker(location=(lat,
    lon)).add_to(my_map)

# Save the map to an HTML file
my_map.save("my_map.html")

# If you're using Jupyter Notebook, you can
    display the map inline
my_map
```



Data Visualization Roles - Change over time



Line chart +Comparisons

Most common chart type for showing change over time. A point is plotted for each time period from left to right; each point's vertical position indicates the feature's value. Points are connected by line segments to emphasize progression across time.



Sparkline +Comparisons

A miniature line chart with little to no labeling, designed to be placed alongside text or in tables. Provides a high-level overview without attracting too much attention. Can also be seen in a sparkbar form, or miniature bar chart (see below).

Data Visualization Roles - Change over time



Connected scatter plot ◆ *+Relationships*

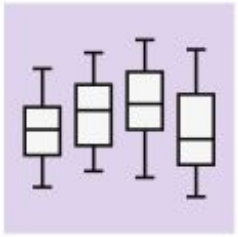
Shows change over time across two numeric variables (see scatter plot in *Relationships*). Line segments still connect points across time, but they may not consistently go from left to right like in a line chart.



Bar chart ● *+Distributions* *+Comparisons*

Each time period is associated with a bar; each bar's value is represented in its height above (or below) a zero-baseline. Works best when there aren't too many time periods to show.

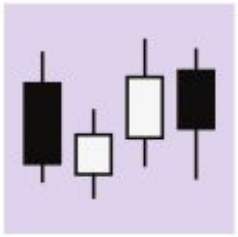
Data Visualization Roles - Change over time



Box plot ■ +Distributions +Comparisons

Each time period is associated with a box and whiskers; each set of box and whiskers shows the range of the most common data values. Best when there are multiple recordings for each time period and a distribution of values needs to be plotted.

Tracking change over time is of key interest in the financial domain. One specialist chart developed for this field includes the following:



Candlestick chart ◆

Looks like a box plot, but each box and whiskers encodes different statistics. The box ends indicate opening and closing prices, while color indicates the direction of change.

Data Visualization Roles - Part-to-whole composition



Pie chart

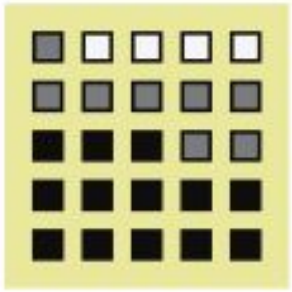
The whole is represented by a filled circle. Parts are proportional slices from that circle, one for each categorical group. Best with five or fewer slices with distinct proportions.



Doughnut chart

A pie chart with a hole in the center. This central area can be used to show a relevant single numeric value. Sometimes used as an aesthetic alternative to a standard progress bar (see stacked bar chart below).

Data Visualization Roles - Part-to-whole composition



Waffle chart / grid plot

Squares laid out in a (typically) 10 x 10 grid; each square represents one percent of the whole. Squares are colored based on categorical group size.



Stacked bar chart

A bar chart (see *Change over time* or *Distributions*) where each bar has been divided into multiple sub-bars to show a part-to-whole breakdown. A single stacked bar can be used as an alternative to the pie or doughnut chart; people tend to make more precise judgments of length over area or angle.

Data Visualization Roles - Part-to-whole composition



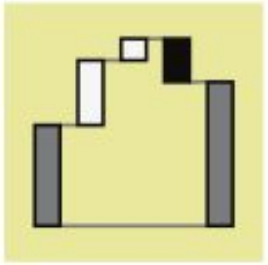
Stacked area chart ●

A line chart (see *Change over time*) where shaded regions are added under the line to divide the total into sub-group values.



Stream graph ◆

Modified version of the stacked area chart where areas are stacked around a central axis. Highlights relative changes instead of exact values.



Waterfall chart ◆

Augments a change over time with a part-to-whole decomposition. Bars on the ends depict values at two time points, and lengths of intermediate floating bars' show the decomposition of the change between points.

Data Visualization Roles - Flows and processes



Funnel chart

Seen in business contexts, showing how people encounter a product and eventually become users or customers. One bar is plotted for each stage, whose lengths reflect the number of users. Connecting regions emphasize connections in stages and give the chart type's namesake shape.



Parallel sets chart

Multiple part-to-whole divisions on different dimensions are depicted as parallel stacked bars. Connecting regions show how different subgroups relate to one another between dimensions.

Data Visualization Roles - Flows and processes



Sankey diagram ◆

The width of the colored region shows the relative volume at each part of a process. Allows for multiple sources of inputs and outputs to be visualized.



Gantt chart ■

Used for project scheduling, breaking them down into individual tasks. Each task is associated with a bar, providing a timeline for when each task should begin and end.

Data Visualization Roles - Geographical data



Scatter map

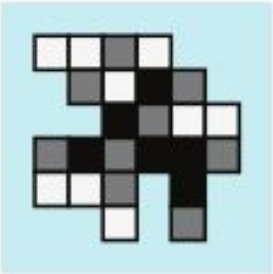
Scatter plot built on top of a geographical map, using geographic coordinates as point positions.



Bubble map

Bubble chart built on top of a geographic map, where point size is an indicator of value. Can also be used to group together points in a scatter map if they are too dense.

Data Visualization Roles - Geographical data



2-d histogram ●

Heatmaps can be built on top of geographic areas. Sometimes seen with a hexagon-shaped grid rather than a rectangular grid. May distort the geography on its edges.



Isopleth / contour map ◆

2-d density curve built on top of a geographic map.



Connection map ◆

Network information and flows built on top of a geographic map.

Data Visualization Roles - Geographical data



Choropleth ●

Similar to a heatmap, but colors are assigned to geopolitical regions rather than an arbitrary grid. Values are often in the form of rates or ratios to avoid distortion due to population density.



Cartogram ◆

Geopolitical regions sized by value. This necessarily requires distortion in shapes and topology.

Machine Learning Pipeline



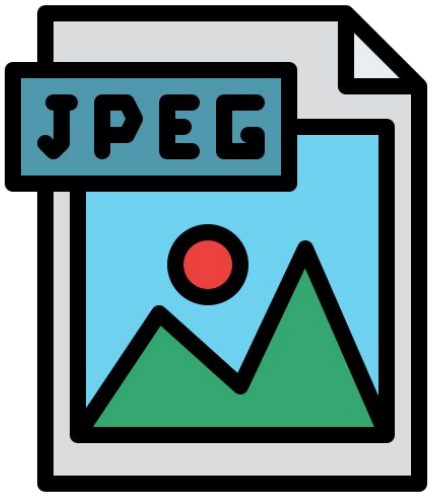
Unstructured Data Type



PDF Document



Paper Form



Image, Photo

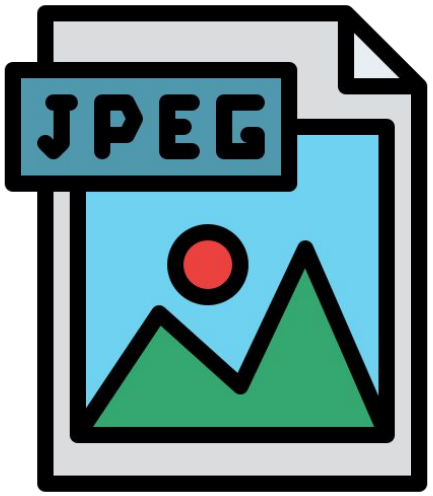
Methodology of Converting Unstructured(Qualitative) to Structured(Quantitative) Data



PDF Document
- pypdf
- pymupdf
- ollama model



Paper Form
- easyocr
- terrasect
- paddleocr



Image, Photo
- easyocr
- terrasect
- paddleocr

Processed Unstructured Data - PDF Document

Step 1 : Open Anaconda Prompt and enter “pip install pymupdf”
Step 2 : Open Jupyter notebook and enter following code

```
# RACAS Application Form Conversion (PDF to form)
def download_pdf(url, save_path):
    response = requests.get(url)
    if response.status_code == 200:
        with open(save_path, 'wb') as f:
            f.write(response.content)
        print(f"PDF downloaded and saved to {save_path}")
    else:
        print(f"Failed to download PDF: {response.status_code}")

import requests
pdf_url =   'https://www.hkag.org/Forms/RACAS%20Form.pdf'
# Usage example
pdf_file_path = "RACAS_applicationform_14012026.pdf"
# Step 1: Download the PDF
download_pdf(pdf_url, pdf_file_path)
```

Processed Unstructured Data - PDF Document

```
import fitz  # PyMuPDF

pdf_file_path = "RACAS_applicationform_14012026.pdf"
pdf_document = fitz.open(pdf_file_path)

# Initialize a list to hold the table data
table_data = []

# Iterate through each page of the PDF
for page_num in range(len(pdf_document)):
    page = pdf_document[page_num]
    text = page.get_text("text")  # Extract text from the page

    # Split the text into lines
    lines = text.splitlines()

    # Process lines to find tables (you may need to adjust this logic)
    for line in lines:
        # Example logic: Split lines by whitespace (this may need
adjustment)
        row = line.split()
        if row:  # If the row is not empty
            table_data.append(row)

pdf_document.close()
# Print extracted table data
for row in table_data:
    print(row)
```

Raw PDF Document



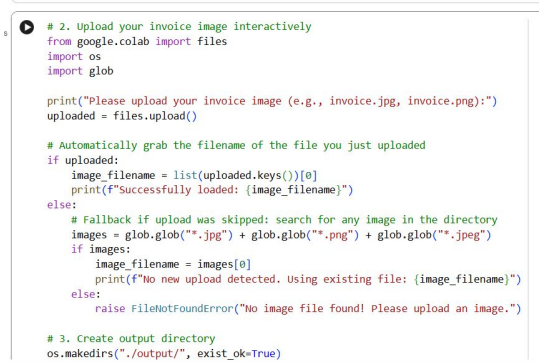
Structured Data

```
['HKAGAS001']
['初版日期:', '04/2005', '上次修定日期：6/2015',
'最新修定日期：1/2020']
['香港老年學會']
['香港安老院舍評審計劃']
['九龍尖沙咀金巴利道35', '號金巴利中心1', '樓/']
['新界葵涌石籬(一)邨石興樓地下18-20', '號']
['電話號碼:', '27273250/', '37052383', '傳真號碼:
', '37052507']
['電郵:', 'racas.adm@hkag.org', '網頁:', 'http://
www.hkag.org']
['香港安老院舍評審計劃申請表格']
['院舍名稱：', '_____
_____（中文）']
```

Python colab code to convert Image to Excel File

Procedure

1. Open https://colab.research.google.com/drive/1QiVEe3NuITFPZIYO9J9tUV5XKgiBfFhg#scrollTo=7RCLYGpn6nTQ

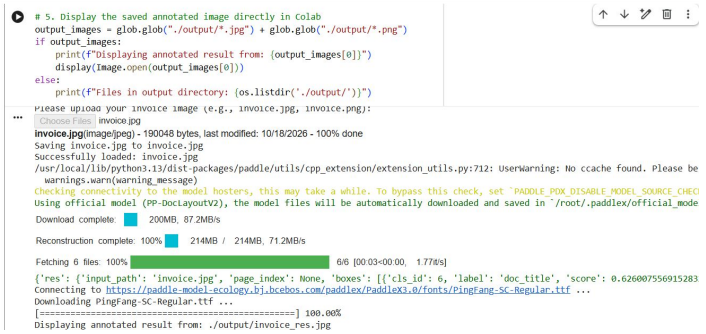


```
# 2. Upload your invoice image interactively
from google.colab import files
import os
import glob

print("Please upload your invoice image (e.g., invoice.jpg, invoice.png):")
uploaded = files.upload()

# Automatically grab the filename of the file you just uploaded
if uploaded:
    image_filename = list(uploaded.keys())[0]
    print(f"Successfully loaded: {image_filename}")
else:
    # Fallback if upload was skipped: search for any image in the directory
    images = glob.glob("*.jpg") + glob.glob("*.png") + glob.glob("*.jpeg")
    if images:
        image_filename = images[0]
        print(f"No new upload detected. Using existing file: {image_filename}")
    else:
        raise FileNotFoundError("No image file found! Please upload an image.")

# 3. Create output directory
os.makedirs("./output/", exist_ok=True)
```



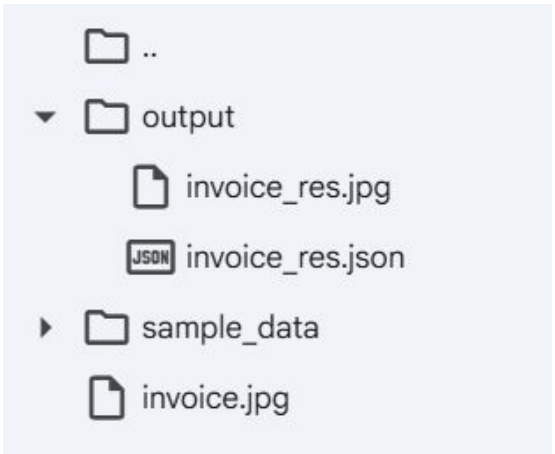
```
# 5. Display the saved annotated image directly in Colab
output_images = glob.glob("./output/*.jpg") + glob.glob("./output/*.png")
if output_images:
    print(f"Displaying annotated result from: {output_images[0]}")
    display(Image.open(output_images[0]))
else:
    print(f"Files in output directory: {os.listdir('./output/')}")
Please upload your invoice image (e.g., invoice.jpg, invoice.png):
Choose Files | invoice.jpg
invoice.jpg(image/jpeg) - 190048 bytes, last modified: 10/18/2026 - 100% done
Saving invoice.jpg to invoice.jpg
Successfully loaded: invoice.jpg
/usr/local/lib/python3.13/dist-packages/paddle/utils/cpp_extension/extension_utils.py:712: UserWarning: No ccache found. Please be
  warnings.warn(warning_message)
Checking connectivity to the model hosters, this may take a while. To bypass this check, set `PADDLE_PDX_DISABLE_MODEL_SOURCE_CHEC
Using official model (PP-DocLayoutV2), the model files will be automatically downloaded and saved in `/root/.paddlex/official_mode
Download complete: 200MB, 87.2MB/s
Reconstruction complete: 100% 214MB / 214MB, 71.2MB/s
Fetching 6 files: 100% 6/6 [00:03<00:00, 1.77it/s]
{'res': {'input_path': 'invoice.jpg', 'page_index': None, 'boxes': [{'cls_id': 6, 'label': 'doc_title', 'score': 0.626007556915283
Connecting to https://paddle-model-ecology.bj.bcebos.com/paddlex/PaddleX3.0/fonts/PingFang-SC-Regular.ttf ...
Downloading PingFang-SC-Regular.ttf ...
[==================================================] 100.00%
Displaying annotated result from: ./output/invoice_res.jpg
```

2. Click “Copy to Drive”
3. Click “Runtime” -> “Run All”
4. Download invoice.jpg from https://github.com/innoviai/IPA_Courses-October2026/blob/main/invoice.jpg
5. User can upload your invoice.jpg
6. Load paddle OCR LLM model to conduct analysis

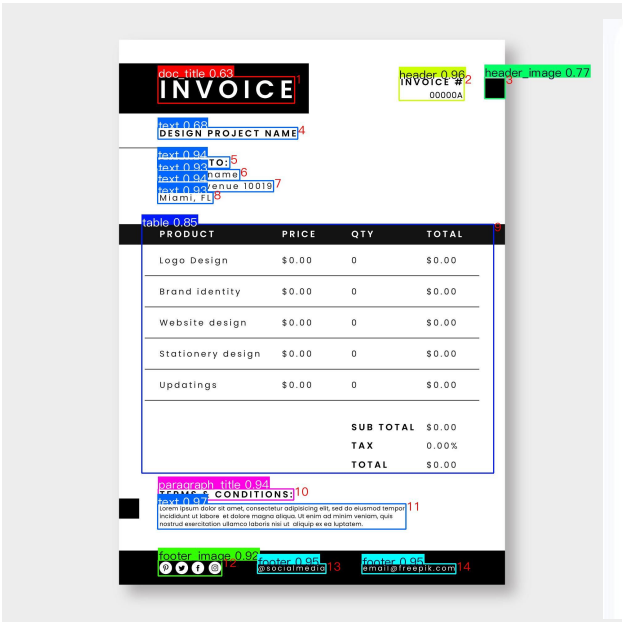
Python colab code to convert Image to Excel File

Procedure

7. An image with labelling and json file is generated.



invoice_res.jpg



invoice_res.json



invoice_res.json


```
{
  "input_path": "invoice.jpg",
  "page_index": null,
  "boxes": [
    {
      "cls_id": 6,
      "label": "doc_title",
      "score": 0.6260075569152832,
      "coordinate": [
        502,
        238,
        944,
        323
      ],
      "order": 1
    },
    {
      "cls_id": 12,
      "label": "header",
      "score": 0.9598553776741028,
      "coordinate": [
        1280,
```

Python anaconda code to convert Image to Excel File

Step 1 : Open Anaconda Prompt and enter “pip install paddleocr paddlepaddle”

Step 2: Open Jupyter notebook and enter following code

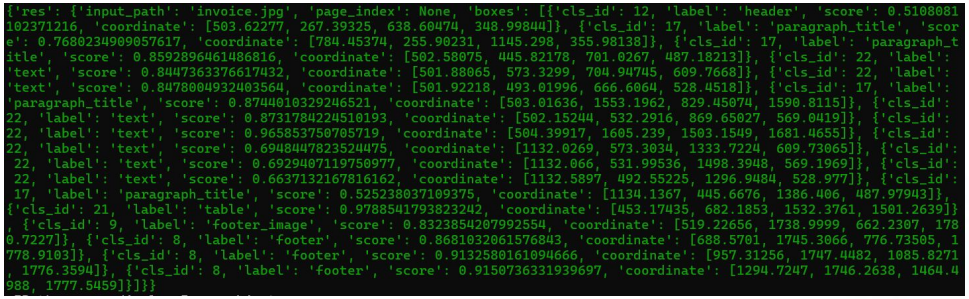
```
#Paddle OCR Layout Analysis Module
#https://www.paddleocr.ai/main/en/version3.x/module_usage/layout_analysis.html#iii-quick-integration
# Initialize the LayoutDetection model
from pathlib import Path
from paddleocr import LayoutDetection

# Input and output paths
image_path = Path(r"C:\Users\Eddie Chow\Downloads\invoice.jpg")
output_dir = Path(r"C:\Users\Eddie Chow\Desktop\hkuspace_lab_oct2026\output")
json_path = output_dir / "invoice_res.json"

# Ensure the output folder exists
output_dir.mkdir(parents=True, exist_ok=True)
# Initialize the layout-detection model
# Disable oneDNN/MKLDNN to avoid the PIR OneDNN conversion error on CPU
model = LayoutDetection(model_name="PP-DocLayoutV2", device="cpu", enable_mkldnn=False,)

# Run layout detection
output = model.predict(input=str(image_path),batch_size=1,layout_nms=True,)

# Process and save each result
for result in output:
    result.print()
    result.save_to_img(save_path=str(output_dir))
    result.save_to_json(save_path=str(json_path))
print(f"Done. Results saved to: {output_dir}")
print(f"JSON result: {json_path}")
```

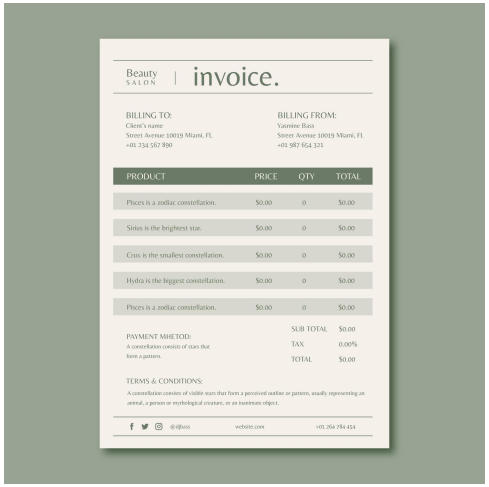


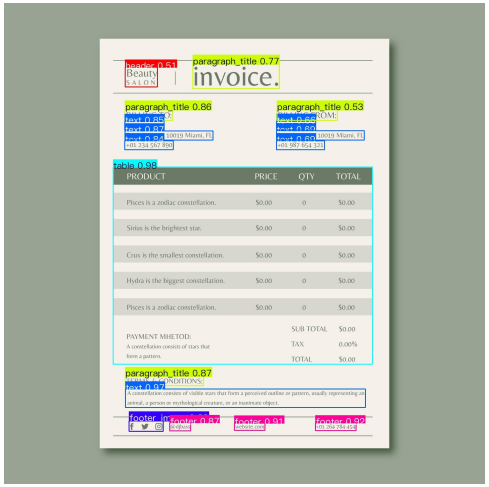
Python anaconda code to convert Image to Excel File

```
#Display original image and labelled image
from IPython.display import Image, display

# Specify the path to your image file
orig_image_path = "invoice.jpg"
label_image_path = "./output/invoice_res.jpg"

# Display the images with specified resolution
display(Image(filename=orig_image_path, width=640, height=480))
display(Image(filename=label_image_path, width=640, height=480))
```

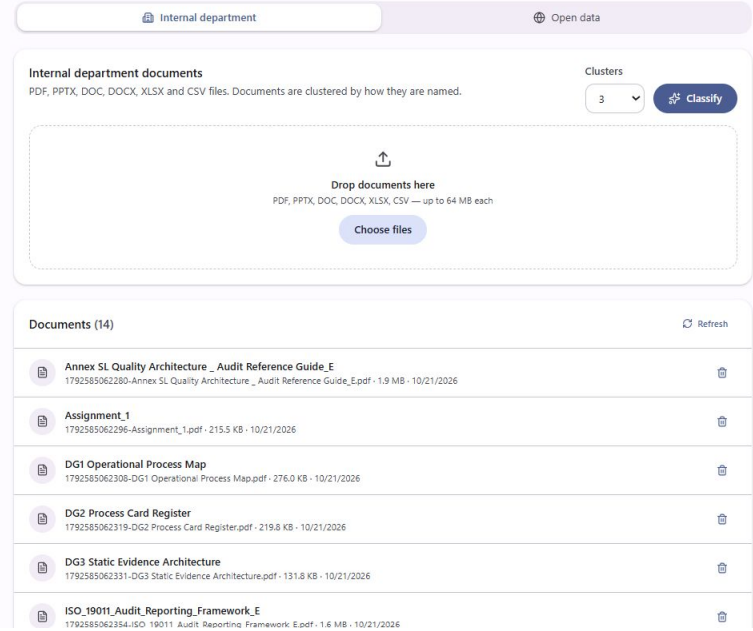




Type of Data Sources

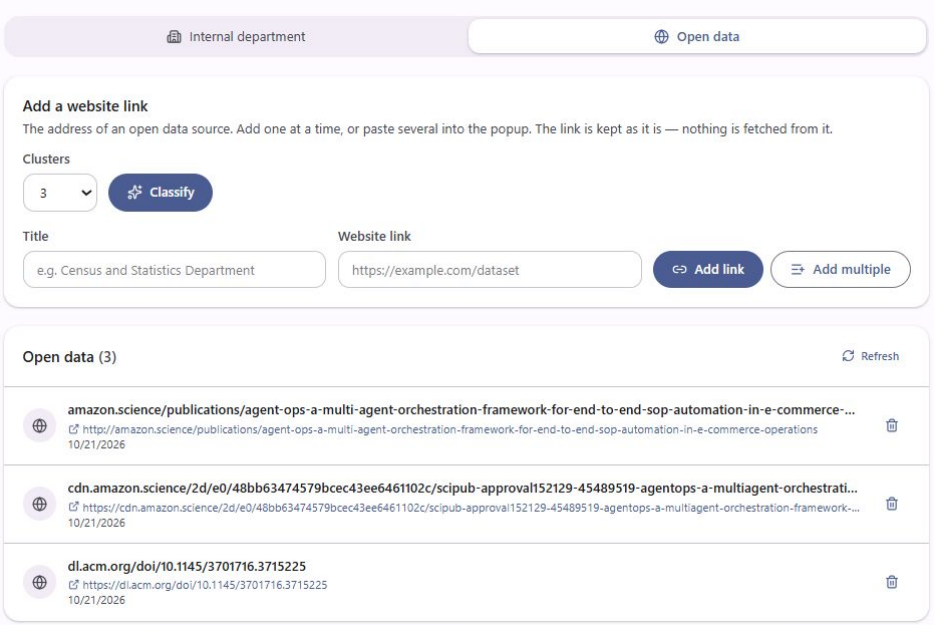
Private Source

- Internal Enterprise Document
- Google Drive, Gmail, Outlook
- IoT Sensor Data



Public Source

- Website URL
- Weather Forecasting
- Satellite Images



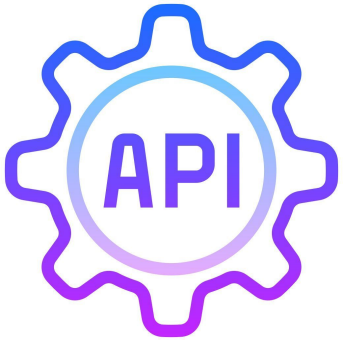
Private Structured Data Collection Method

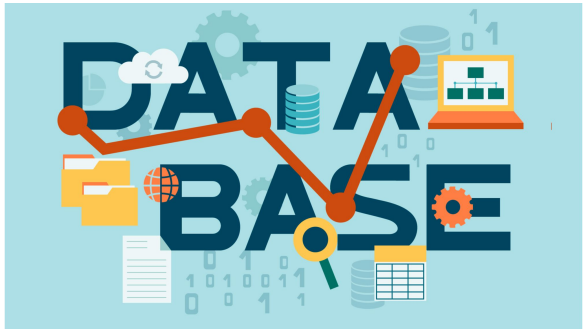
Main Data collection are get from Google Sheets, API, Web Scraping, Gmail, Database











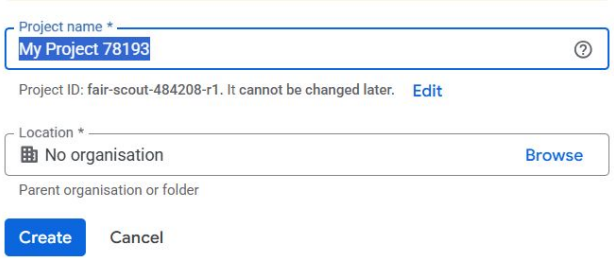
Data Collection - Reading Data from Gmail

Enable the Google Sheets API with read permission.

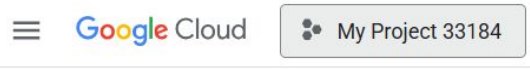
1. Open Browser and enter https://console.cloud.google.com/
2. Click “Select a project”
3. Click “New Project”



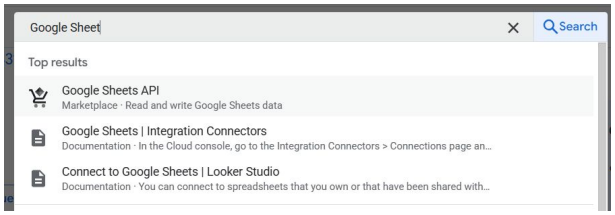
4. Enter information and click “Create”



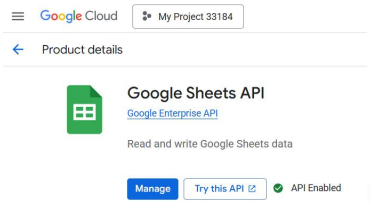
5. Select your Project that create



6. in Search Bar, enter ‘Google Sheet’



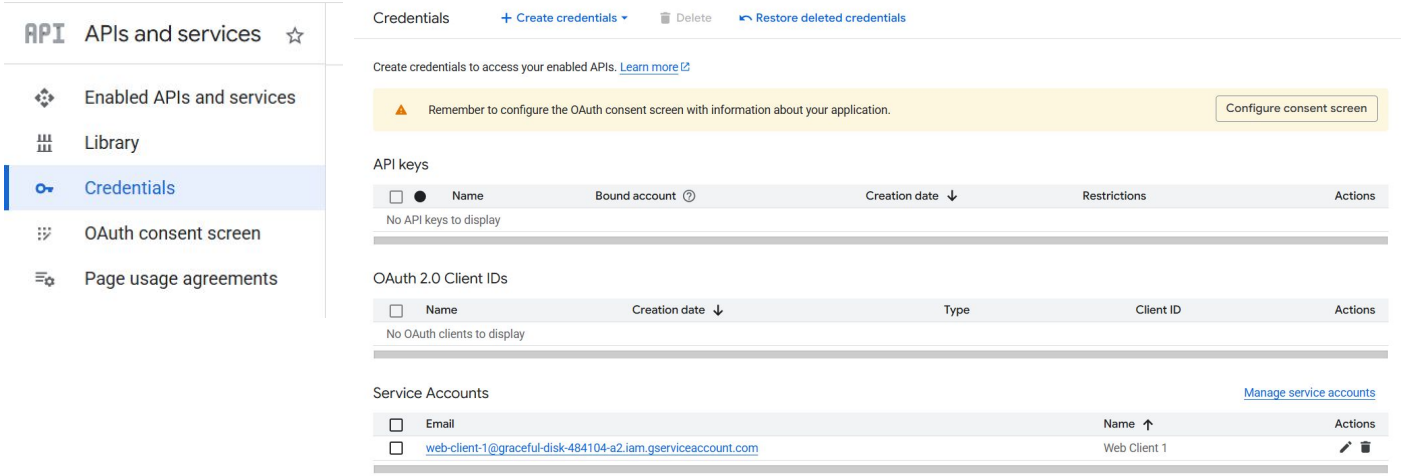
7. Select “Gmail API”
8. Click “Enable”



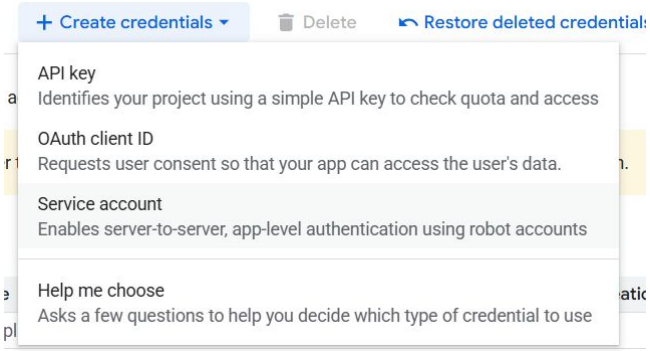
Data Collection - Reading Data from Gmail

Credentials Setting

1. Click Credentials and whole screen is displayed



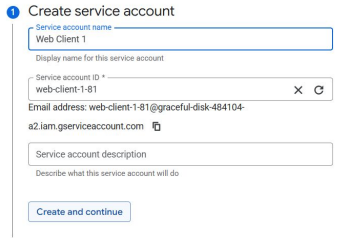
2. Click “Create credentials” and “Service Account”



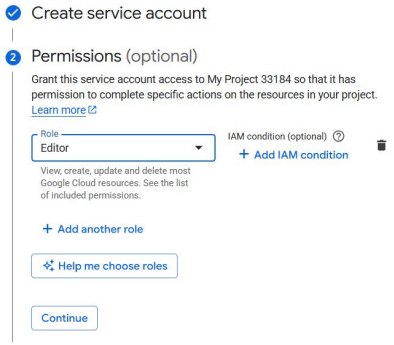
Data Collection - Reading Data from Gmail

Credentials Setting

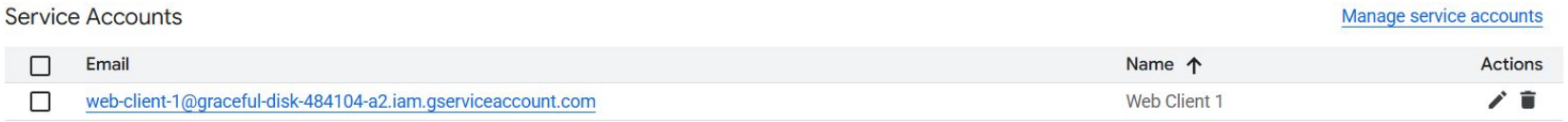
3. In ”Service account name”, enter “Web Client 1” and Click “Create and Continue”



4. In “Select a role”, enter “Editor”, click “Continue”



5. Click “Done” to set credentials.

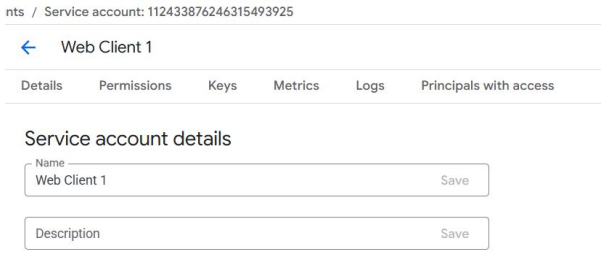


6. Copy the email like “web-client-1@graceful-disk-484104-a2.iam.gserviceaccount.com”

Data Collection - Reading Data from Gmail

Credentials Setting

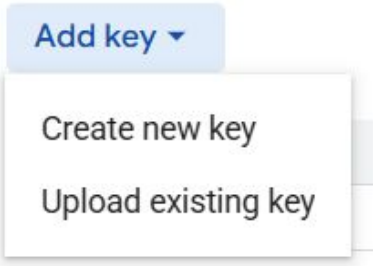
7. Click “Service Account Email”



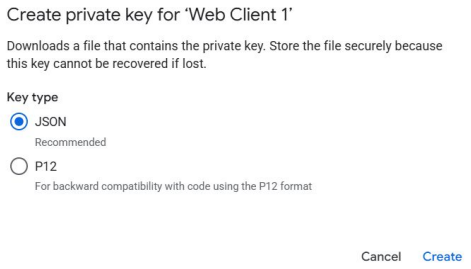
8. Click “Keys”



9. Click “Add Key” and “Create New Key”



10. Enter “JSON”,



11. A JSON Key “credentials.json” is saved in your local PC

Data Collection - Reading Data from Gmail

Conduct Python Code Running to fetch gmail content

1. Open Anaconda Prompt, enter "`pip install beautifulsoup4 google-api-python-client google-auth-httplib2 google-auth-oauthlib`" to install google and web scraping library

```
(base) C:\Users\Eddie Chow>pip install google-api-core google-auth google-api-python-client
Defaulting to user installation because normal site-packages is not writeable
WARNING: Ignoring invalid distribution -ransformers (c:\users\eddie chow\appdata\roaming\python\python39\site-packages)
WARNING: Ignoring invalid distribution -ransformers (c:\users\eddie chow\appdata\roaming\python\python39\site-packages)
Requirement already satisfied: google-api-core in c:\users\eddie chow\appdata\roaming\python\python39\site-packages (1.3
4.1)
Requirement already satisfied: google-auth in c:\users\eddie chow\appdata\roaming\python\python39\site-packages (1.35.0)
Requirement already satisfied: google-api-python-client in c:\users\eddie chow\appdata\roaming\python\python39\site-pack
ages (2.184.0)
Requirement already satisfied: googleapis-common-protos<2.0dev,>=1.56.2 in c:\users\eddie chow\appdata\roaming\python\py
thon39\site-packages (from google-api-core) (1.70.0)
Requirement already satisfied: protobuf!=3.20.0,!=3.20.1,!=4.21.0,!=4.21.1,!=4.21.2,!=4.21.3,!=4.21.4,!=4.21.5,<4.0.0dev
,>=3.19.5 in c:\users\eddie chow\appdata\roaming\python\python39\site-packages (from google-api-core) (3.20.3)
Requirement already satisfied: requests<3.0.0dev,>=2.18.0 in c:\users\eddie chow\appdata\roaming\python\python39\site-pa
ckages (from google-api-core) (2.32.5)
Requirement already satisfied: cachetools<5.0,>=2.0.0 in c:\programdata\anaconda3\lib\site-packages (from google-auth) (
4.2.2)
Requirement already satisfied: pyasn1-modules>=0.2.1 in c:\programdata\anaconda3\lib\site-packages (from google-auth) (0
.2.8)
Requirement already satisfied: setuptools>=40.3.0 in c:\programdata\anaconda3\lib\site-packages (from google-auth) (61.2
.0)
Requirement already satisfied: six>=1.9.0 in c:\programdata\anaconda3\lib\site-packages (from google-auth) (1.16.0)
Requirement already satisfied: rsa<5,>=3.1.4 in c:\programdata\anaconda3\lib\site-packages (from google-auth) (4.7.2)
Requirement already satisfied: httplib2<1.0.0,>=0.19.0 in c:\users\eddie chow\appdata\roaming\python\python39\site-packa
ges (from google-api-python-client) (0.31.0)
Requirement already satisfied: google-auth-httplib2<1.0.0,>=0.2.0 in c:\users\eddie chow\appdata\roaming\python\python39
\site-packages (from google-api-python-client) (0.2.0)
```

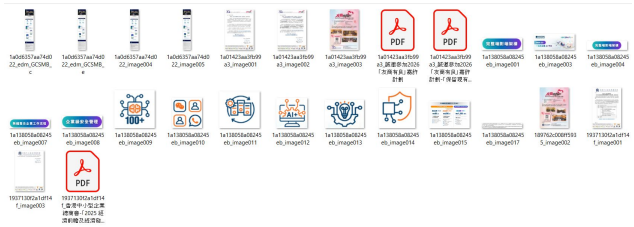
Data Collection - Reading Data from Gmail

Conduct Python Code Running to fetch gmail content

2. Download code at https://github.com/innoviai/IPA_Courses-October2026/blob/main/single_read_gmail_csvcontent_fromsubjectname.py
3. Open single_read_gmail_csvcontent_fromsubjectname.py code
4. Edit the target_subject value in line 117.

```
117     target_subject = "香港中小型企業總商會"
118     query = f'subject:"{target_subject}"'
119
120     results = service.users().messages().list(userId='me', q=query).execute()
121     messages = results.get('messages', [])
```

5. Open anaconda prompt to run code by entering "python single_read_gmail_csvcontent_fromsubjectname.py".
6. An email.csv, attachments folder and token.pickle file is created.



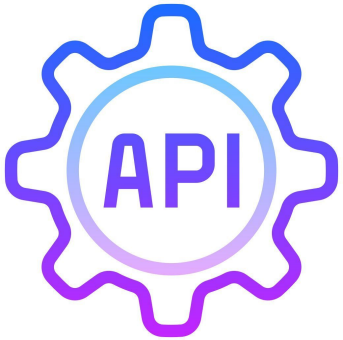
Private Structured Data Collection Method

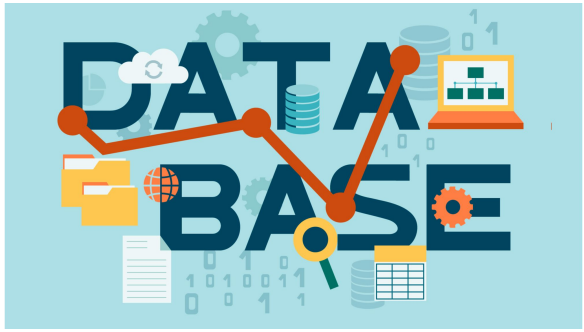
Main Data collection are get from Google Sheets, API, Web Scraping, Gmail, Database











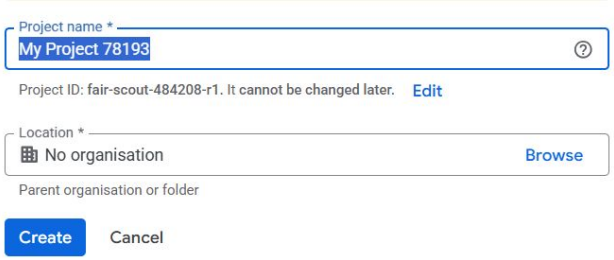
Data Collection - Reading Weather Data from Google Sheet

Enable the Google Sheets API with readpermission.

1. Open Browser and enter https://console.cloud.google.com/
2. Click “Select a project”
3. Click “New Project”



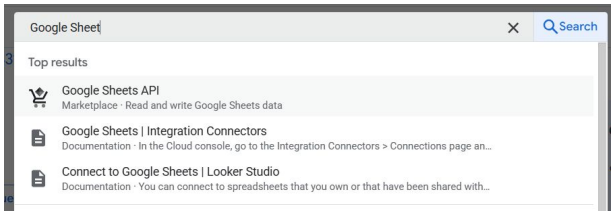
4. Enter information and click “Create”



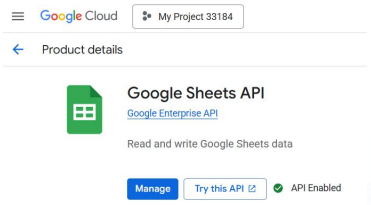
5. Select your Project that create



6. in Search Bar, enter ‘Google Sheet’



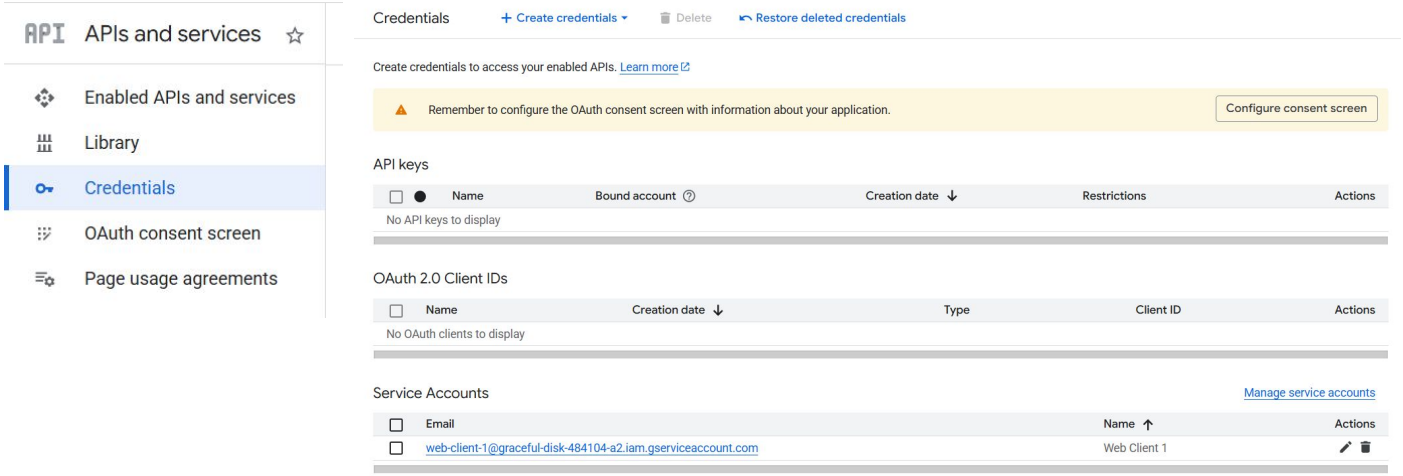
7. Select “Google Sheet API”
8. Click “Enable”



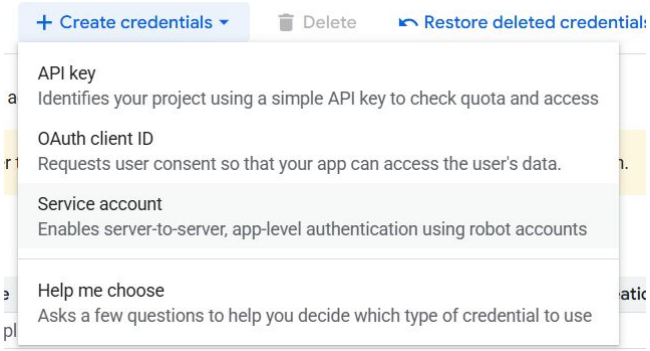
Data Collection - Reading Weather Data from Google Sheet

Credentials Setting

1. Click Credentials and whole screen is displayed



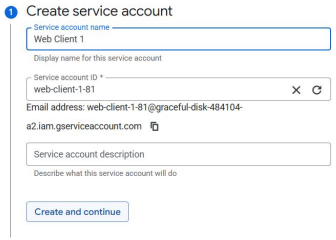
2. Click “Create credentials” and “Service Account”



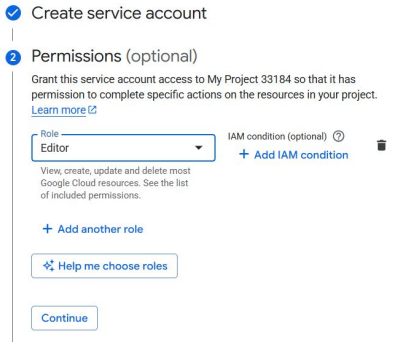
Data Collection - Reading Weather Data from Google Sheet

Credentials Setting

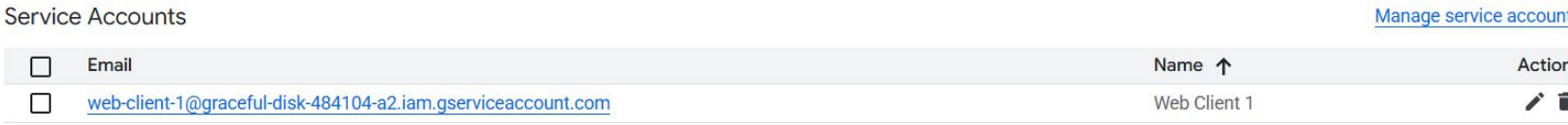
3. In ”Service account name”, enter “Web Client 1” and Click “Create and Continue”



4. In “Select a role”, enter “Editor”, click “Continue”



5. Click “Done” to set credentials.

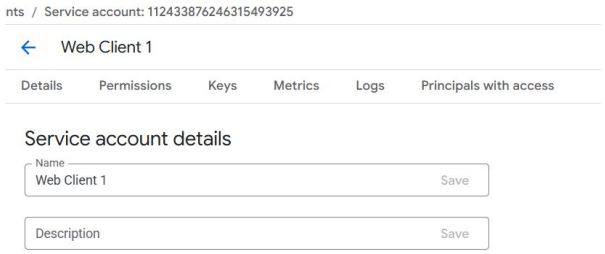


6. Copy the email like “web-client-1@graceful-disk-484104-a2.iam.gserviceaccount.com”

Data Collection - Reading Weather Data from Google Sheet

Credentials Setting

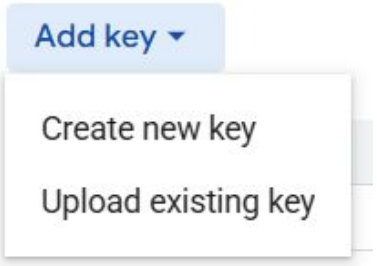
8. Click “Service Account Email”



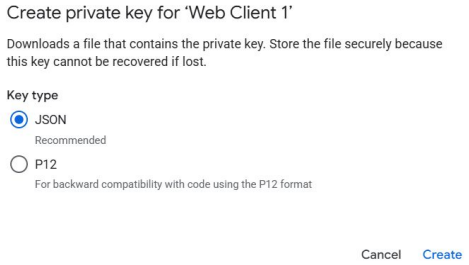
9. Click “Keys”



10. Click “Add Key” and “Create New Key”



11. Enter “JSON”,

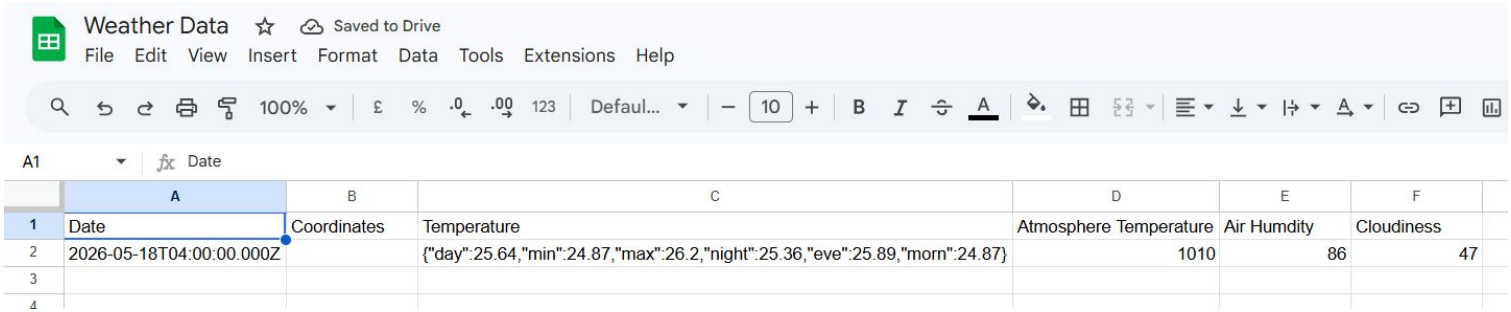


12. A JSON Key is saved in your local PC

Data Collection - Reading Weather Data from Google Sheet

Google Sheet Settings

1. Open previous weather data google sheet



	A	B	C	D	E	F
1	Date	Coordinates	Temperature	Atmosphere Temperature	Air Humdity	Cloudiness
2	2026-05-18T04:00:00.000Z		{"day":25.64,"min":24.87,"max":26.2,"night":25.36,"eve":25.89,"morn":24.87}	1010	86	47
3						
4						

1. Click “File”, Share, “Share with Others”
2. Enter “web-client-1@graceful-disk-484104-a2.iam.gserviceaccount.com”
3. Make sure it is in editor, click “Send”
4. From the google sheet website url like “https://docs.google.com/spreadsheets/d/1BVdJkKlZQPTdGrzL-h7ymTYT6unhYKgJ68vqqUYsjP0/edit?gid=0#gid=0”
5. Copy sheet_id **“1BVdJkKlZQPTdGrzL-h7ymTYT6unhYKgJ68vqqUYsjP0”** , range_name ”**Sheet1!A1:F2** “ in notepad

Data Collection - Reading Weather Data from Google Sheet

Python Coding

1. Open Anaconda Prompt, enter "`pip install google-api-core google-auth google-api-python-client google-auth-httplib2 google-auth-oauthlib`" to install google library

```
(base) C:\Users\Eddie Chow>pip install google-api-core google-auth google-api-python-client
Defaulting to user installation because normal site-packages is not writeable
WARNING: Ignoring invalid distribution -ransformers (c:\users\eddie chow\appdata\roaming\python\python39\site-packages)
WARNING: Ignoring invalid distribution -ransformers (c:\users\eddie chow\appdata\roaming\python\python39\site-packages)
Requirement already satisfied: google-api-core in c:\users\eddie chow\appdata\roaming\python\python39\site-packages (1.3
4.1)
Requirement already satisfied: google-auth in c:\users\eddie chow\appdata\roaming\python\python39\site-packages (1.35.0)
Requirement already satisfied: google-api-python-client in c:\users\eddie chow\appdata\roaming\python\python39\site-pack
ages (2.184.0)
Requirement already satisfied: googleapis-common-protos<2.0dev,>=1.56.2 in c:\users\eddie chow\appdata\roaming\python\py
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Requirement already satisfied: protobuf!=3.20.0,!=3.20.1,!=4.21.0,!=4.21.1,!=4.21.2,!=4.21.3,!=4.21.4,!=4.21.5,<4.0.0dev
,>=3.19.5 in c:\users\eddie chow\appdata\roaming\python\python39\site-packages (from google-api-core) (3.20.3)
Requirement already satisfied: requests<3.0.0dev,>=2.18.0 in c:\users\eddie chow\appdata\roaming\python\python39\site-pa
ckages (from google-api-core) (2.32.5)
Requirement already satisfied: cachetools<5.0,>=2.0.0 in c:\programdata\anaconda3\lib\site-packages (from google-auth) (
4.2.2)
Requirement already satisfied: pyasn1-modules>=0.2.1 in c:\programdata\anaconda3\lib\site-packages (from google-auth) (0
.2.8)
Requirement already satisfied: setuptools>=40.3.0 in c:\programdata\anaconda3\lib\site-packages (from google-auth) (61.2
.0)
Requirement already satisfied: six>=1.9.0 in c:\programdata\anaconda3\lib\site-packages (from google-auth) (1.16.0)
Requirement already satisfied: rsa<5,>=3.1.4 in c:\programdata\anaconda3\lib\site-packages (from google-auth) (4.7.2)
Requirement already satisfied: httplib2<1.0.0,>=0.19.0 in c:\users\eddie chow\appdata\roaming\python\python39\site-packa
ges (from google-api-python-client) (0.31.0)
Requirement already satisfied: google-auth-httplib2<1.0.0,>=0.2.0 in c:\users\eddie chow\appdata\roaming\python\python39
\site-packages (from google-api-python-client) (0.2.0)
```

Data Collection - Reading Weather Data from Google Sheet

Python Coding

2. Open jupyter notebook
3. Type following code

```
# Import Library
import os
import csv
from google.oauth2.service_account import Credentials
from googleapiclient.discovery import build

SCOPES = ['https://www.googleapis.com/auth/spreadsheets']
SERVICES_ACCOUNT_FILE = "graceful-disk-484104-a2-152ebd9c692e.json"

credentials = Credentials.from_service_account_file(SERVICES_ACCOUNT_FILE, scopes=SCOPES)
service = build('sheets', 'v4', credentials=credentials)

sheet_id = '1BVdJkKlZQPTdGrzL-h7ymTYT6unhYKgJ68vqqUYsjP0'
range_name = "'Sheet1'!A1:F2"  # Specify the range with the sheet name
```

Data Collection - Reading Weather Data from Google Sheet

```
# Retrieve the data
result = service.spreadsheets().values().get(spreadsheetId=sheet_id, range=range_name).execute()
values = result.get('values', [])
print(values)
```

```
[['Date', 'Coordinates', 'Temperature', 'Atmosphere Temperature', 'Air Humdity', 'Cloudiness'], ['2026-05-18T04:00:00.000Z', '', '{"day":25.64,"min":24.87,"max":26.2,"night":25.36,"eve":25.89,"morn":24.87}', '1010', '86', '47']]
```

```
if not values:
        print('No data found.')
else:
        # Save to CSV file
        with open('weatherforecastdata.csv', 'w', newline='', encoding='utf-8') as csv_file:
                writer = csv.writer(csv_file)
                writer.writerows(values)  # Write all the rows

        print('Data has been saved to weatherforecastdata.csv')
```

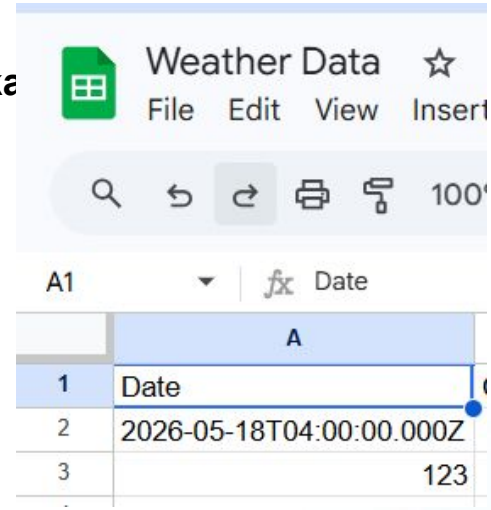
(Optional) Google Sheet Content Update

```
# Update the range to just the starting cell for appending
 result = service.spreadsheets().values().update(
     spreadsheetId=sheet_id,
     range=range_name,  # Change range here
     valueInputOption=value_input_option,
     body=body
 ).execute()

 print(f"{result.get('updatedCells')} cells updated.")
 return result
```

sheet_id = '1mqrxumbU1Si5aGmkg2jpffu7zn9uWxFfvXX1Wnuwka
range_name = "Sheet1!A3"
value_input_option = 'USER_ENTERED'
values = [['123']]

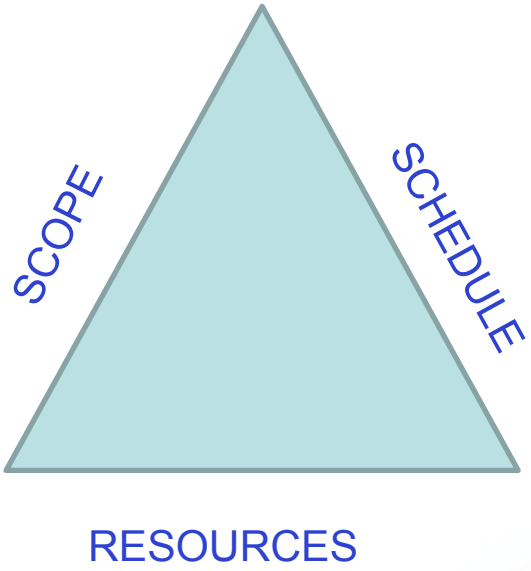
update_values(sheet_id, range_name, value_input_option, values)



What is a Project?

Unique process consisting of a set of coordinated and controlled activities with start and finish dates, undertaken to achieve an objective conforming to specific requirements, including constraints of time, cost, quality and resources

1. A Project is a planned set of activities
2. A Project has a scope
3. A Project has time, cost, quality and resource constraints



Project Management

What is Project Management?

Project Management is the process of planning, organizing, and overseeing tasks and resources to achieve a specific goal within a set timeline, budget, and scope. It ensures projects are completed efficiently, on time, and meet objectives.

Why It Matters:

- Keeps teams organized & aligned
- Manages time, cost & resources effectively
- Reduces risk & improves quality of outcome

Key Types of Project Management



Waterfall

Linear, step-by-step approach. Best for projects with clear requirements.



Agile

Flexible & iterative. Ideal for evolving projects & fast delivery.



Scrum

Framework under Agile. Uses sprints to deliver work in cycles.



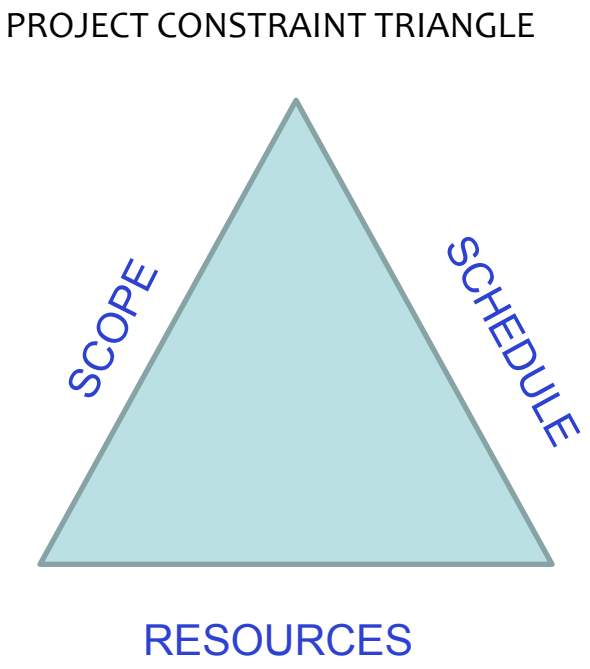
Kanban

Visual workflow. Focuses on continuous delivery & task flow.

Project Management The “Triple Constraint”

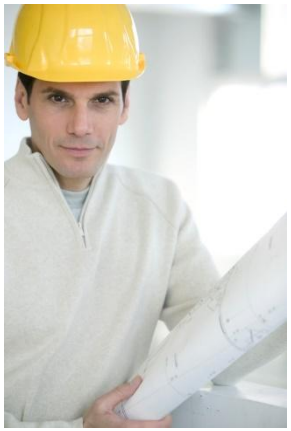
The Project Manager / the researcher has certain constraints to battle with in delivering a project. These are referred to as “The Triple Constraint”. They include:

- Scope Management
- Cost Management
- Time Management
- Quality Assurance
- Human Resource Management
- Communication Management
- Risk Management
- Conflict Management among others.



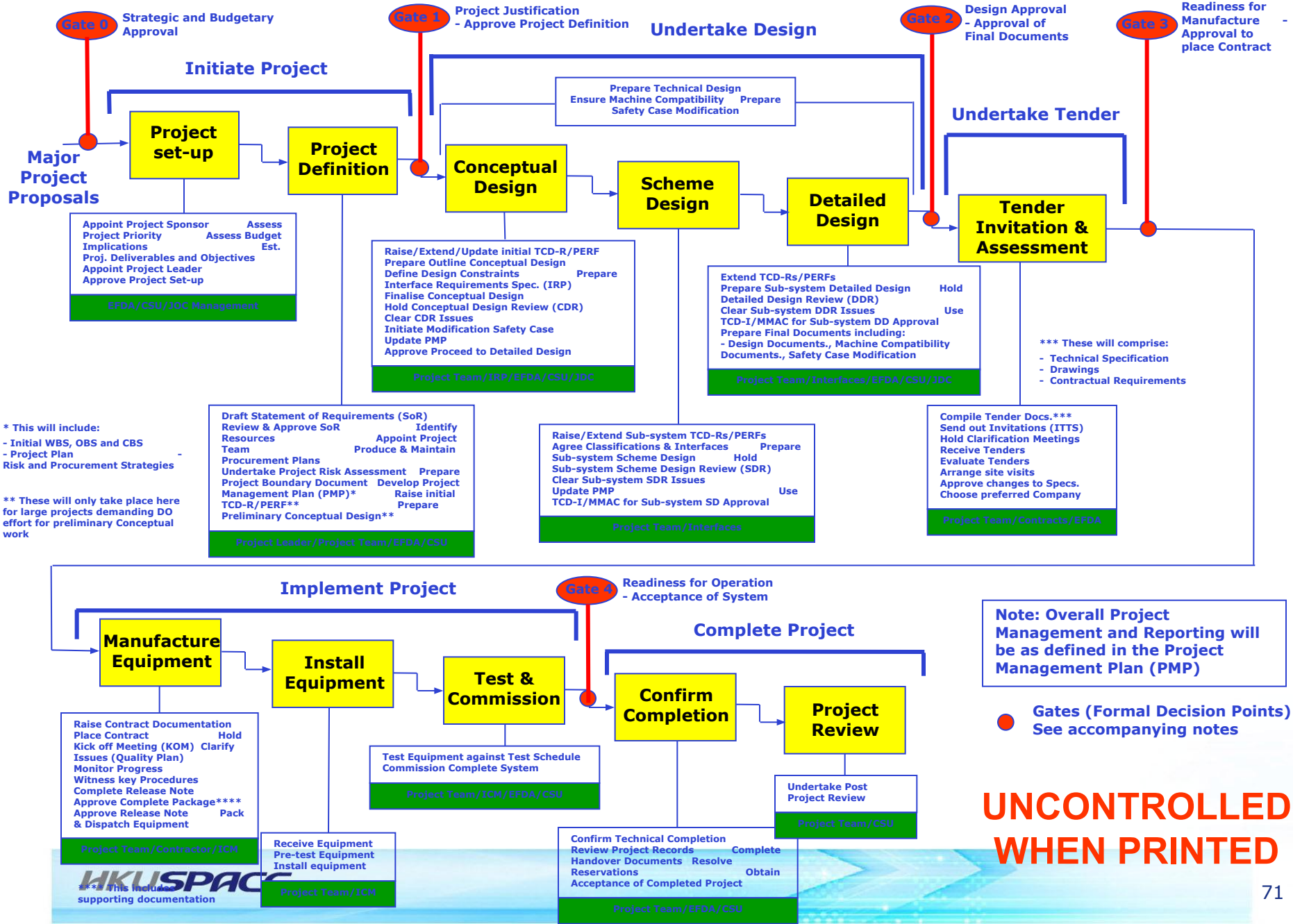
Project Manager Role

A Good Project Manager



- Takes ownership of the whole project
- Is proactive not reactive
- Adequately plans the project
- Is Authoritative (**NOT** Authoritarian)
- Is Decisive
- Is a Good Communicator
- Manages by data and facts not uniformed optimism
- Leads by example
- Has sound Judgement
- Is a Motivator
- Is Diplomatic
- Can Delegate

The Project Process

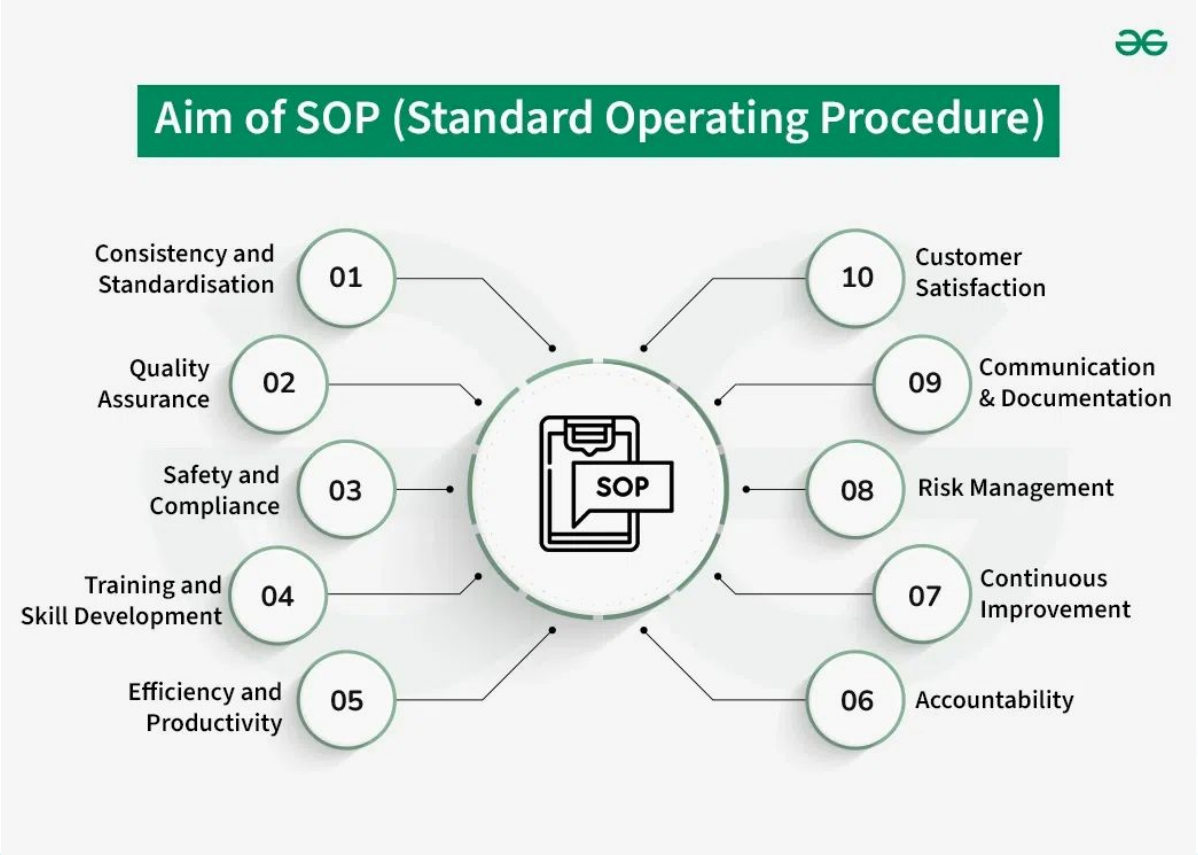


UNCONTROLLED WHEN PRINTED

What is SOP(Standard Operational Procedures)?

SOPs define the standard way work should be done

An **SOP (Standard Operating Procedure)** is a reusable, documented set of instructions for *how* to perform a recurring task consistently. In short: **the project defines what must be achieved and when; SOPs define how the work is done.**



AI Tools (Copilot) to generate SOP with standard data

Conduct Business Research on several global benchmark or certificate such as NEC Contract, HKEX ESG Reporting, Beam Plus

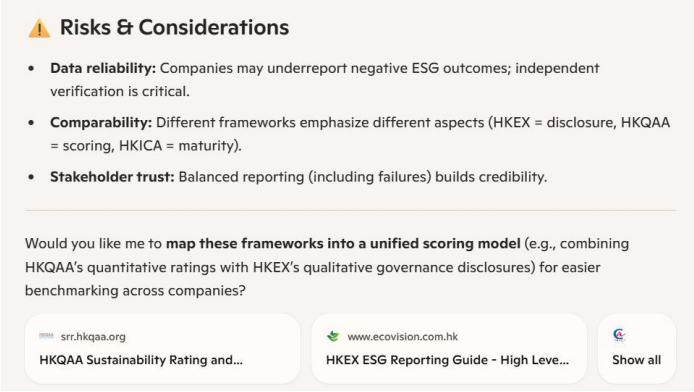
Flow
1. Open copilot at https://copilot.microsoft.com/
2. Enter prompt “conduct HK ESG research that provide assessment methodology, scoring criteria, final outcomes in quantitative or qualitative approach.”

… you can based on above research to find other bench-marking and conduct your AI transformation journey



Comparison of ESG Approaches in Hong Kong

Framework	Methodology	Scoring Criteria	Outcome
HKQAA Rating	Plan-Do-Check-Act + accountability	Company data, public research, risk ratings	Quantitative scores for Hang Seng Index inclusion
HKEX Guide	Disclosure principles (Materiality, Quantitative, Balance, Consistency)	KPIs + governance disclosures	Qualitative + quantitative ESG reports
HKICA Certification	Maturity model + auditor evaluation	KPIs, leadership, culture, disclosure	Certificate + maturity rating



Risks & Considerations

- **Data reliability:** Companies may underreport negative ESG outcomes; independent verification is critical.
- **Comparability:** Different frameworks emphasize different aspects (HKEX = disclosure, HKQAA = scoring, HKICA = maturity).
- **Stakeholder trust:** Balanced reporting (including failures) builds credibility.

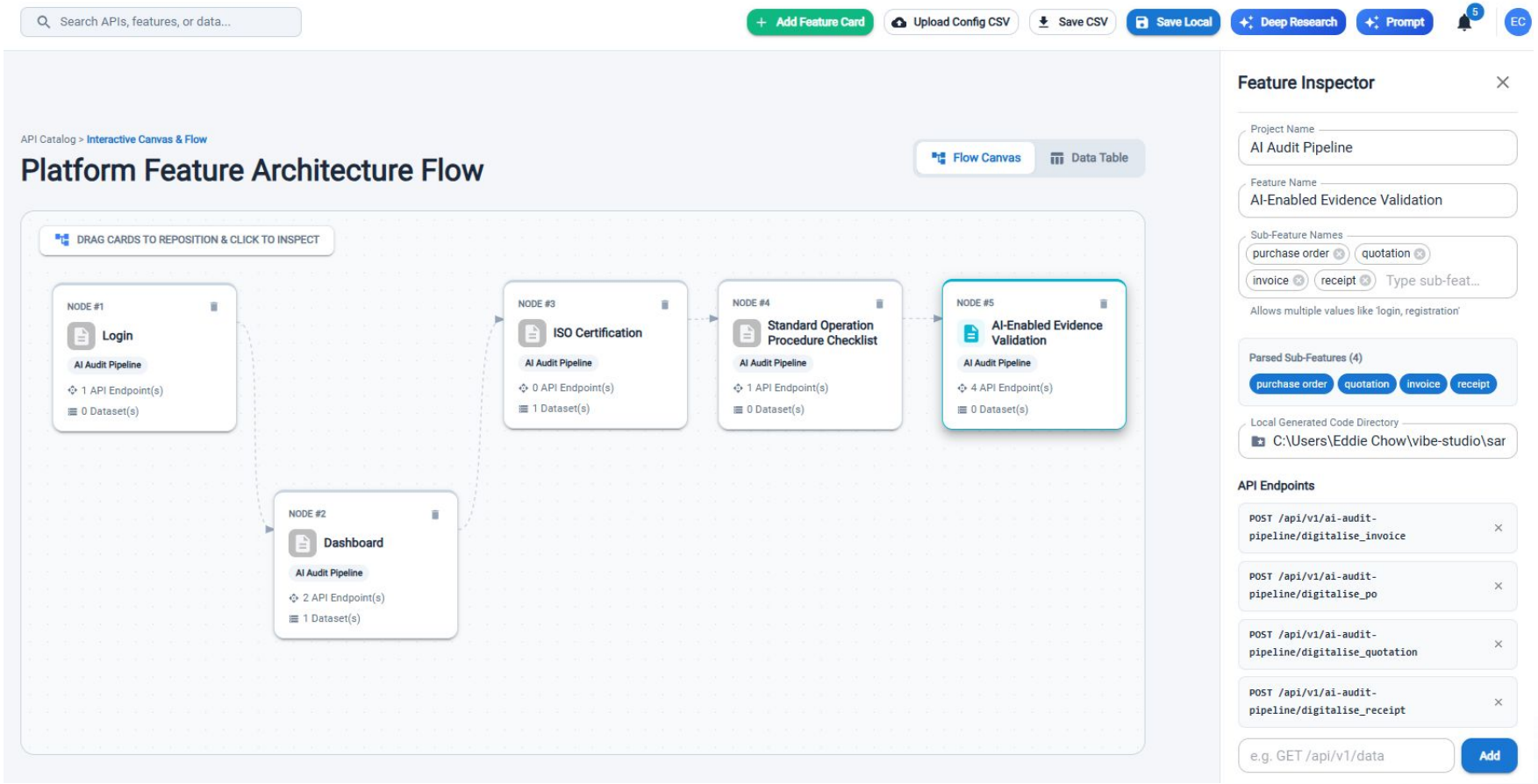
Would you like me to **map these frameworks into a unified scoring model** (e.g., combining HKQAA’s quantitative ratings with HKEX’s qualitative governance disclosures) for easier benchmarking across companies?

srr.hkqaa.org
HKQAA Sustainability Rating and...

www.ecovision.com.hk
HKEX ESG Reporting Guide - High Leve...

Show all

SOP Design Flow Chart



Key Points in Project Set-up and Definition

- Create Project Management Plan (PMP)
- Be clear of scope and objectives
- Establish clear statement of what is to be done (WBS)
- Establish Risks to be Managed
- Establish Costs and Durations
- Establish Resources Required

Project management Plan - PMP

- Master Document for Project
- Defines the following:-
  - Project Objectives, Scope, Deliverables
  - Stakeholders (Internal & External)
  - Work to be done (WBS)
  - Project Organisation and Resources (OBS)
  - Project Costings (CBS)
  - Project Schedule
  - Procurement/Contract Strategy
  - Risk Management
  - Quality management
  - Change Management

Project Planning

Adequate planning leads to the correct completion of work



Plan

Inadequate planning leads to frustration towards the end of the project & poor project performance





Project Start

Project End

Work Breakdown Structure (WBS)

The Work Breakdown Structure is the foundation for effective project planning, costing and management.

It is the most important aspect in setting-up a Project

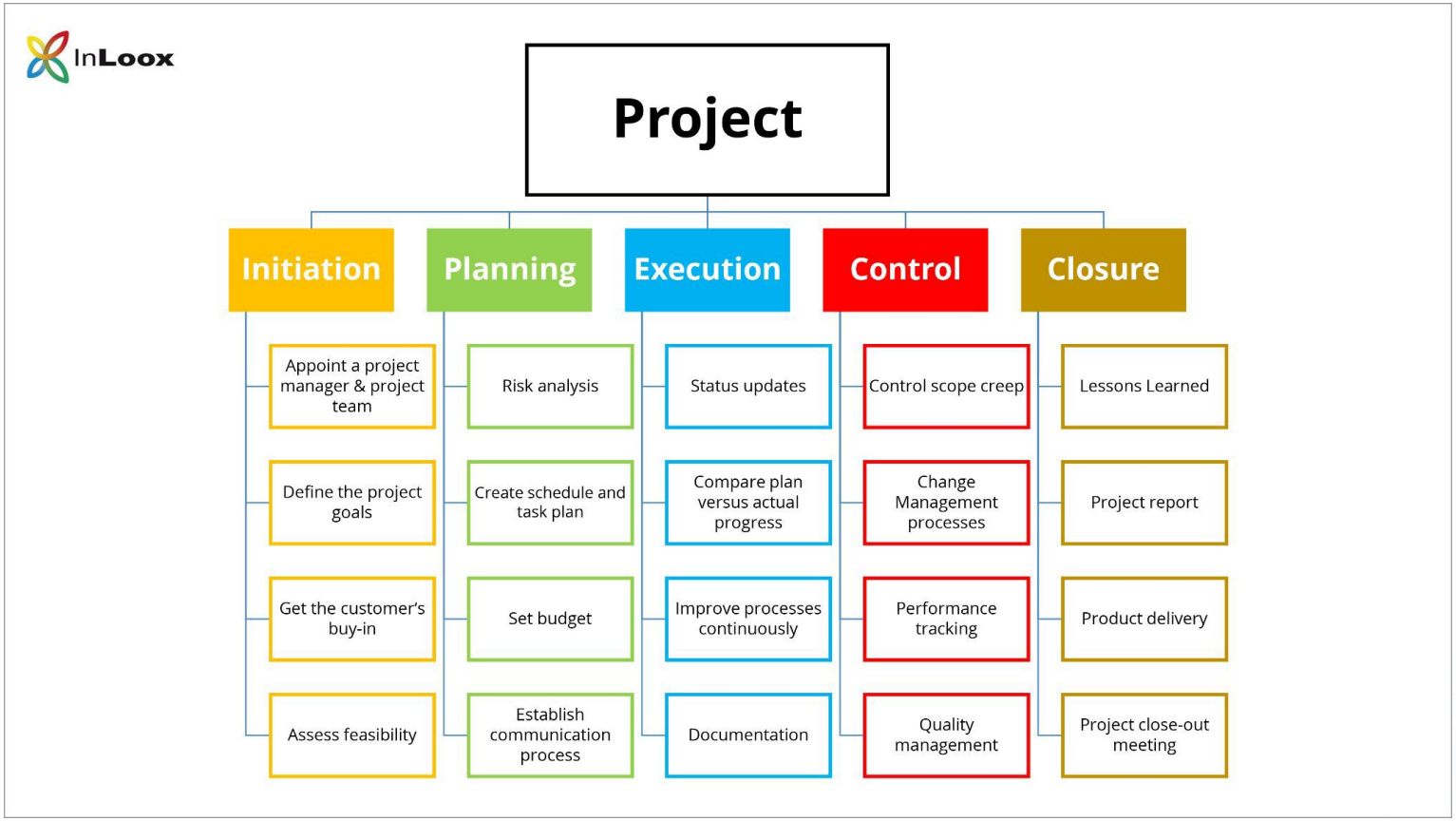
- It is the foundation on which everything else builds



Work Breakdown Structure - Definition

"A Work Breakdown Structure (WBS) is a hierarchical (from general to specific) tree structure of deliverables and tasks that need to be performed to complete a project."

Example WBS - Top Level ILW Project

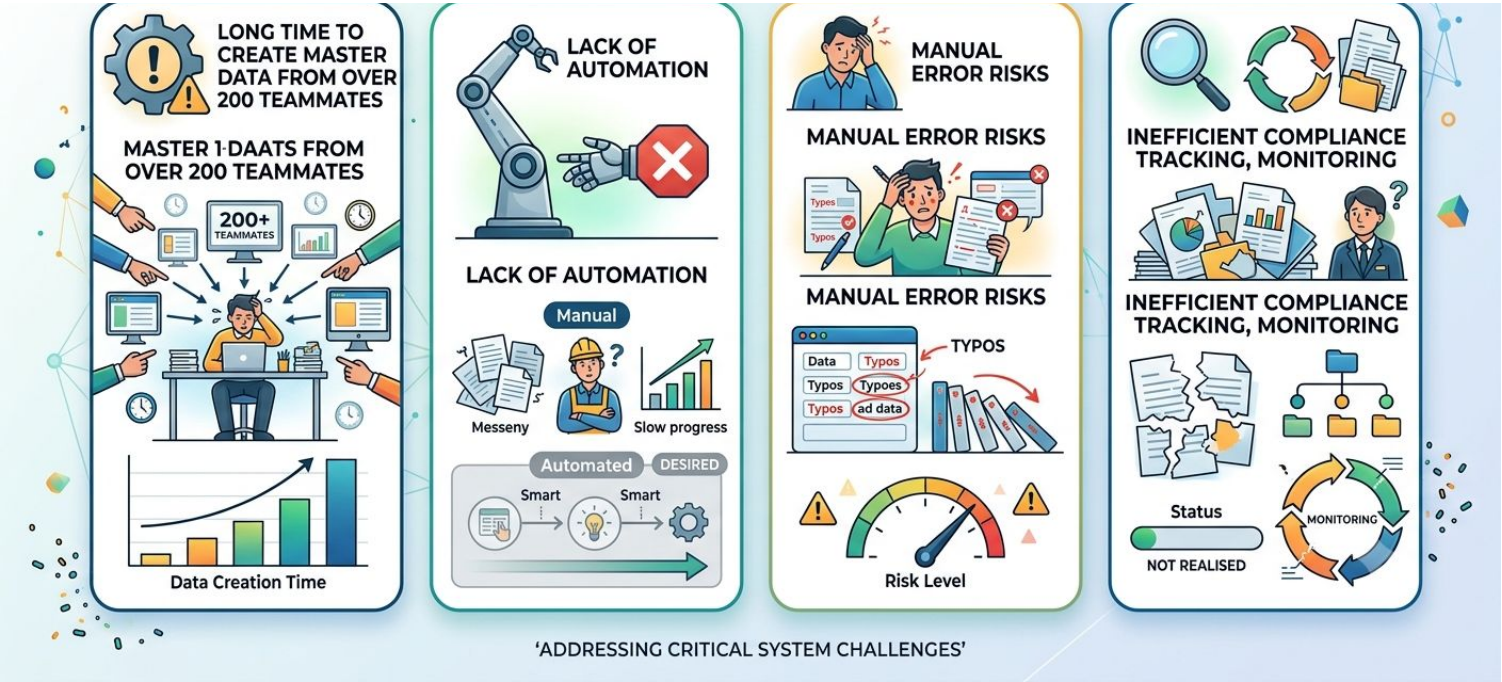


Example WBS - Top Level TSCL Project



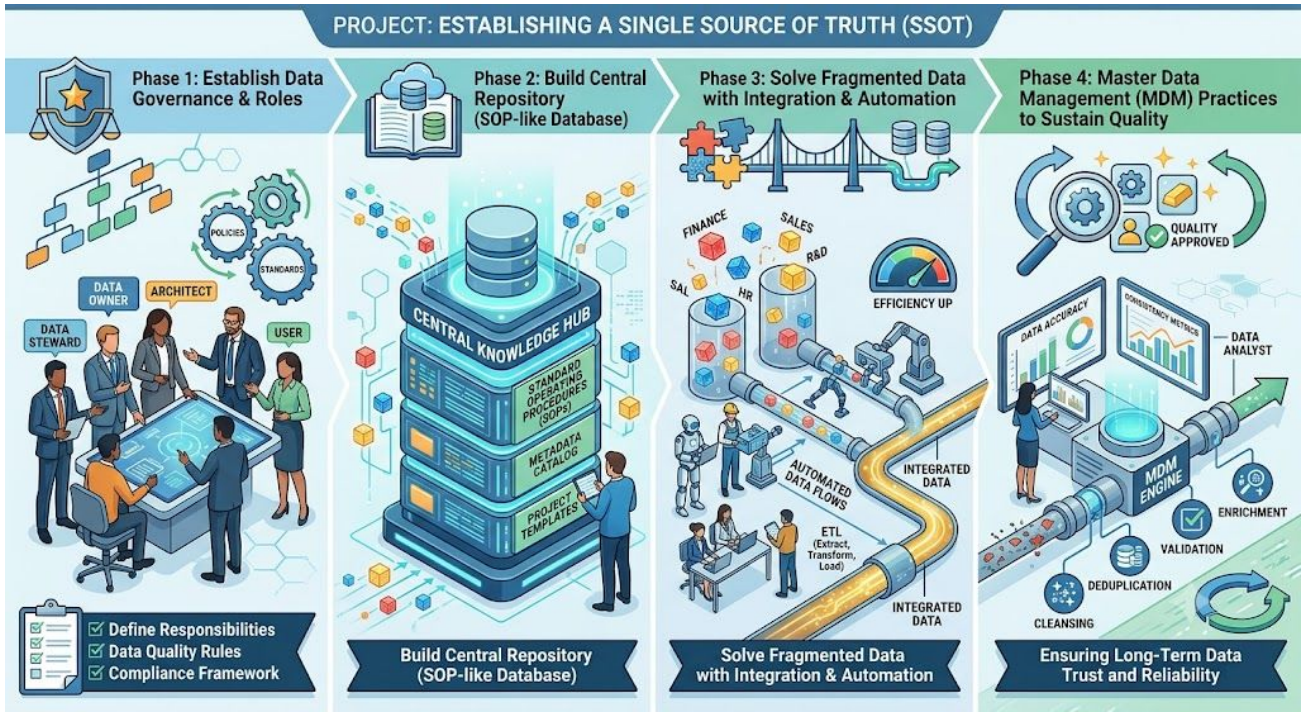
Company Data Pains Points

- Long time to process data collection from online and offline data
- Long time to create master data from over 200 teammates
- Lack of automation
- Manual error risks
- Inefficient compliance tracking, monitoring



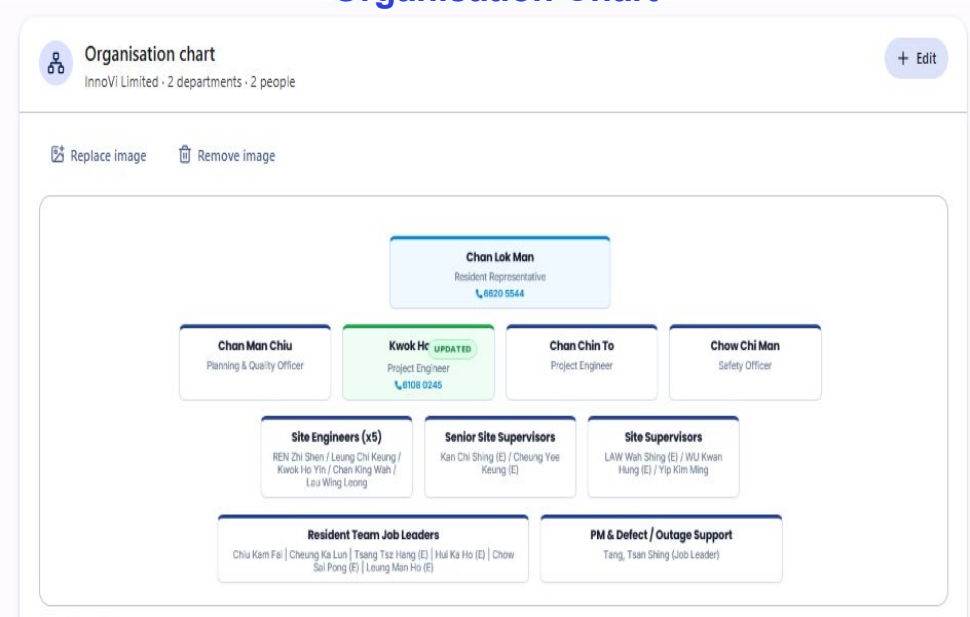
Solutions

- Establish a “single source of truth” based on different project that used for aligning diverse data source
- Implement data governance and roles
- Build the central repository (SOP-like database)
- Solve fragmented data with integration and automation
- Master Data Management practices to sustain quality

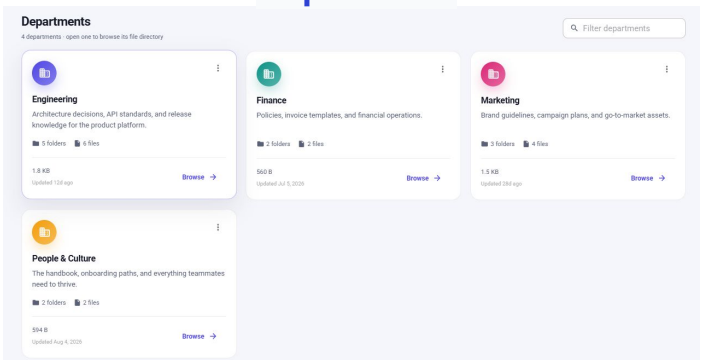


Build Company OS Knowledge

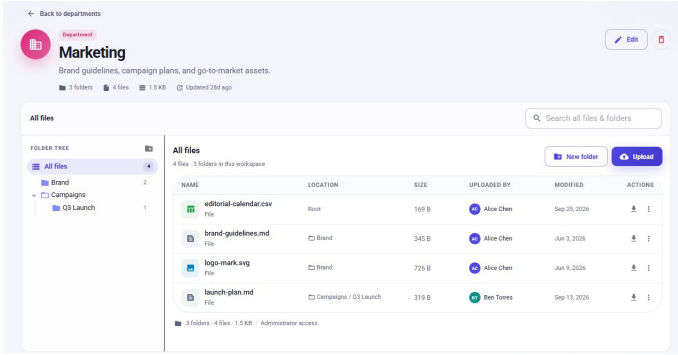
Organisation Chart



Department

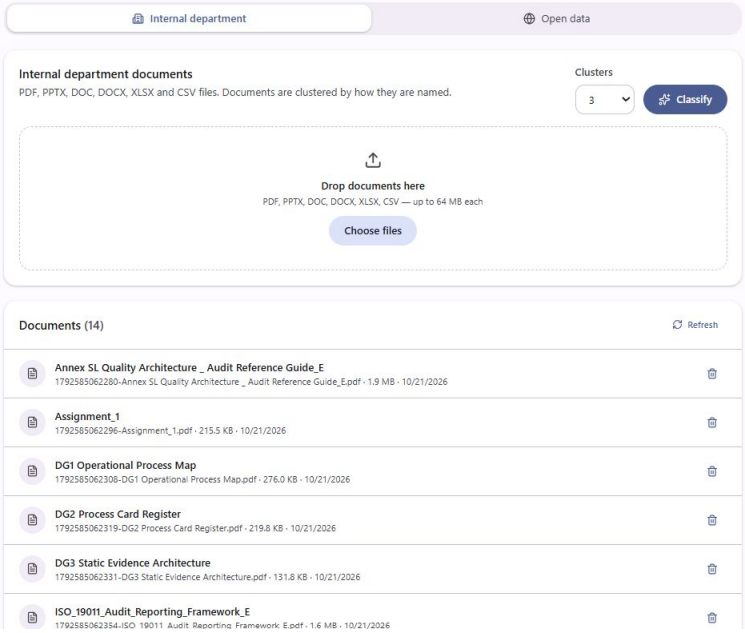


FIle Structure

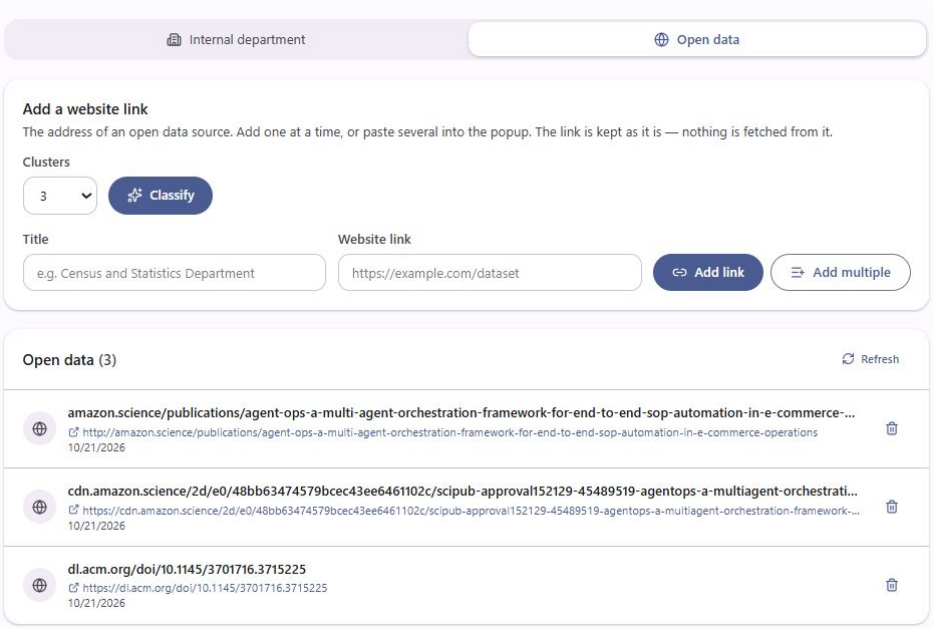


Build Company OS Knowledge Types

Enterprise Internal Document

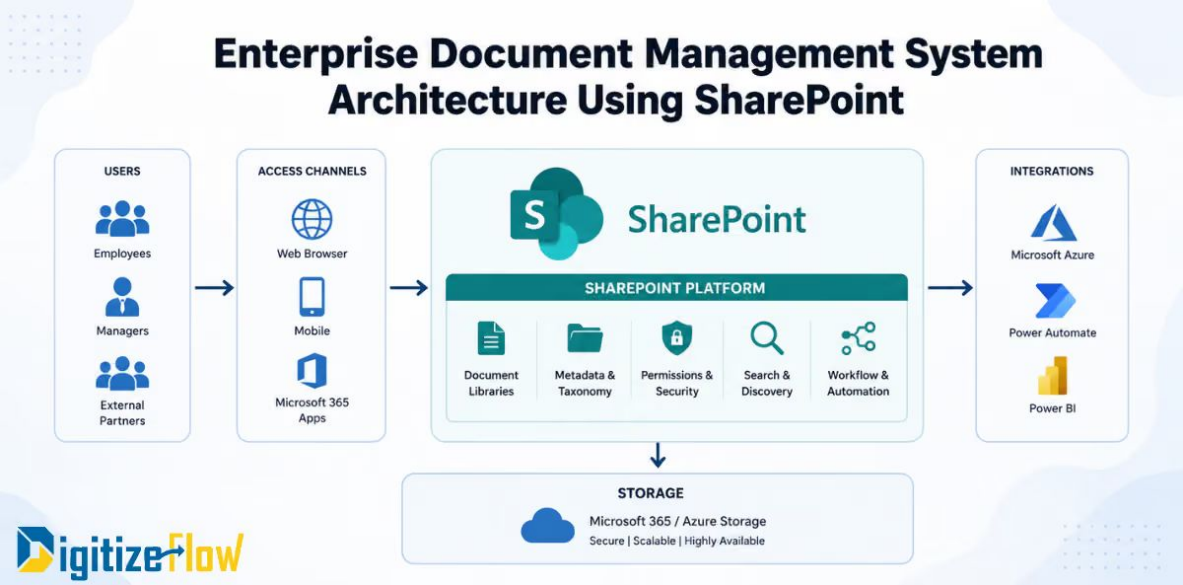


Open Data Knowledge



Enterprise Document Management System Example

SharePoint powers a modern Enterprise Document Management System (EDMS) with secure document storage, intelligent search, workflow automation, metadata management, and seamless Microsoft 365 integration.



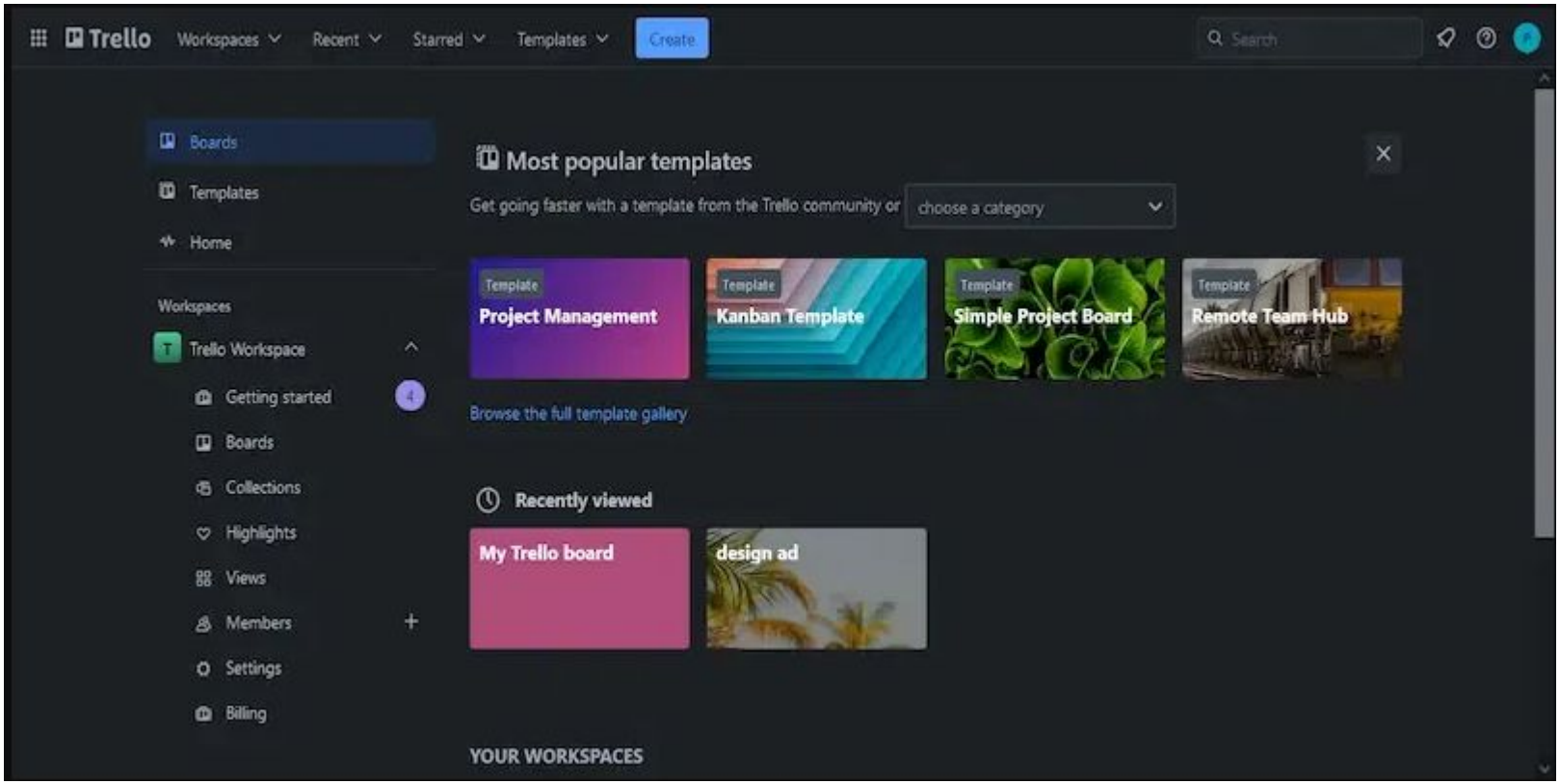
https://www.digitizeflow.com/blog/enterprise-document-management-system-architecture

Tools for doing Project Management Plan

1. Gantt Charts: visualize timelines, track progress, and manage tasks effectively.
2. Asana: Offers flexible task management with various visualization options like lists, Kanban boards, and Gantt charts
3. Trello: A visual tool that organizes projects into boards and cards, ideal for agile workflows
4. Notion: Maintaining an internal database, collaborating, and managing tasks such as Creating project plans, Building roadmaps, Creating and storing important documents, guides, etc.
5. Monday.com : monday.com is a highly visual and intuitive project management tool that offers a range of customization options.
6. Jiro : Jira is amongst the most established, powerful, and feature-complete project management tools available. It is straightforward to set up and available both on the cloud and on premises.
7. Wrike: Features task assignment, collaboration tools, and multiple viewing formats (list, board, Gantt) to manage workflows efficiently

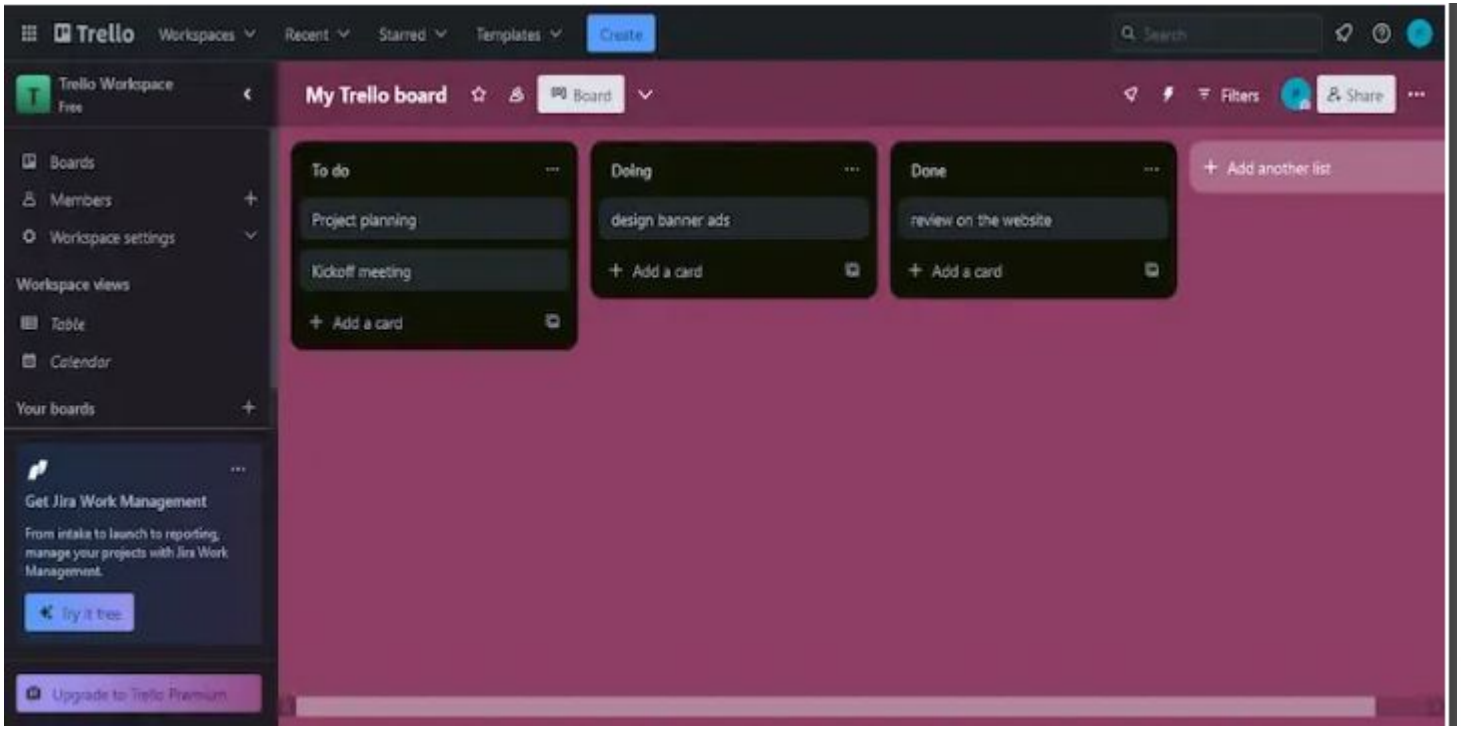
What is Trello?

Trello is a task and project management tool that's comprised of Boards, Lists, and Cards.



Trello Board

Trello boards are the advanced version of bulletin boards that are used to organize thoughts. The specialty of Trello is that you are not restricted to using a single board and can use multiple boards as complex projects will require multiple boards. It also provides different privacy options: **personal**, **private**, **team**, **organization**, and **public visibility**. Your board will show you **what is planned to be done**, and **task statuses,** and help you indicate capacity limitations.

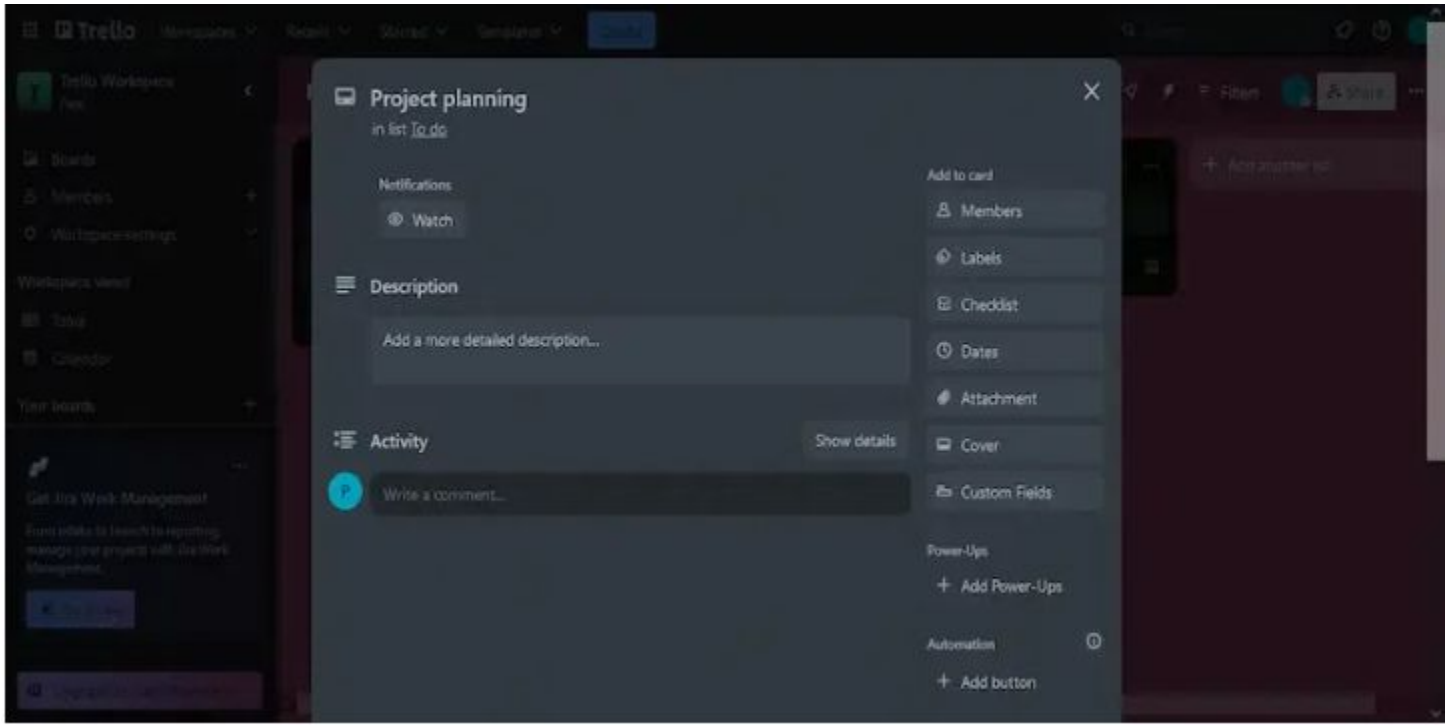


Exercise 1

1. **Sign Up and Create an Account**: Go to Trello's website and sign up for a free account.
2. **Create a Board**: Click on the "**Create new board**" button and give your board a name.
3. **Add Lists**: Inside your board, create lists to represent different stages of your workflow (e.g., To Do, In Progress, Done).
4. **Create Cards**: Add cards under each list for individual tasks. Click on a card to add details, due dates, attachments, and comments.
5. **Collaborate**: Invite team members by clicking on the "**Invite**" button and assigning them to specific cards.
6. **Move Cards**: Drag and drop cards between lists to reflect the progress of tasks.
7. **Use Power-Ups**: Enhance your board with **Power-Ups** (integrations and additional features) for added functionality.

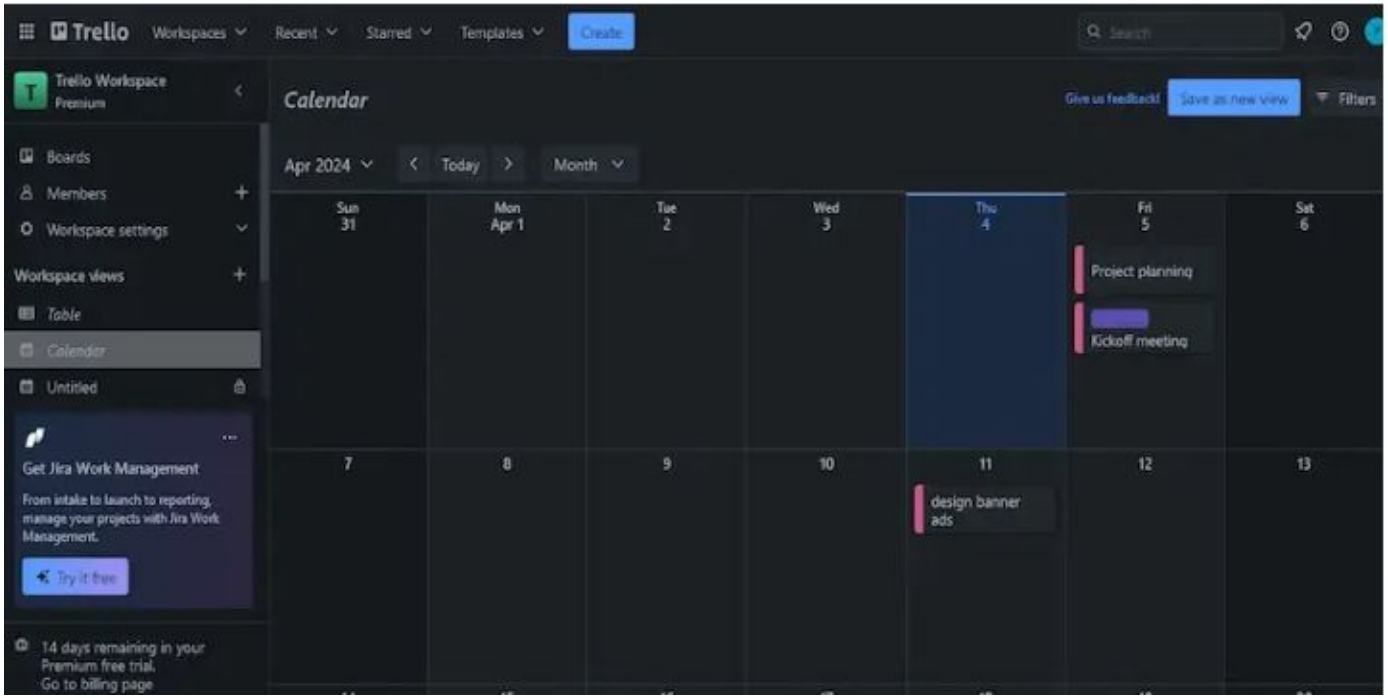
Using Trello Cards

Once the lists on the board are **completed**, cards need to be added. These Cards are the **building block of Trello** which provides details of the tasks within the project. Clicking a card expands to show detailed information like the example shown below.



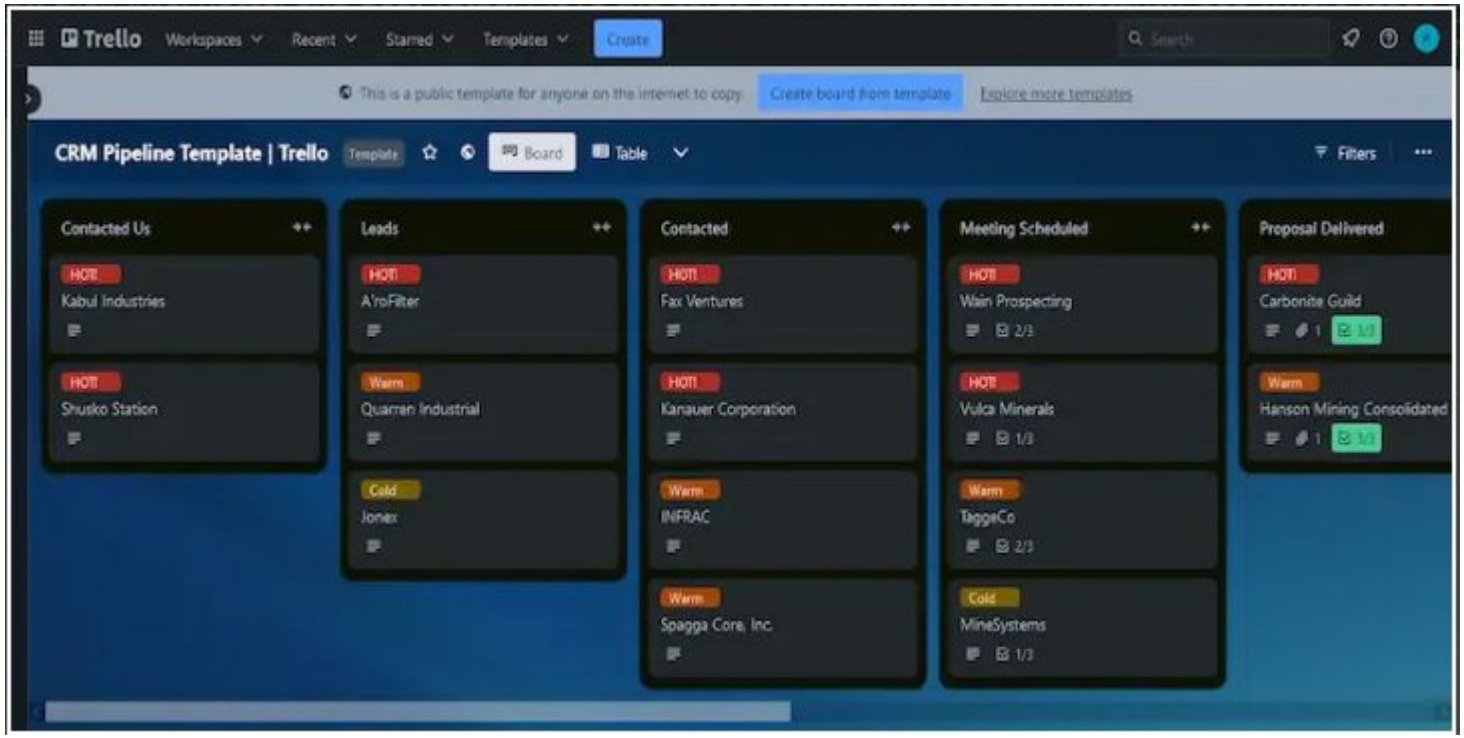
Calendar Planning

Trello's built-in view makes it a calendar planner, allowing you to see your deadlines visually, organized by due date. Lists act as timeframes ("**This Week**"), while color-coded labels highlight priorities. Optional Power-Ups further enhance functionality, letting you sync with external calendars. This makes Trello a powerful yet flexible way to manage your time, no matter how complex your schedule gets. By leveraging these features, you can effectively use Trello for project management, ensuring that all tasks are tracked and completed on time.



CRM Tool

Trello can be an essential tool in managing Customer relationships. Each customer has a Trello card, similar to a digital folder holding all their info, notes, and deal value. Move the cards around, and you can instantly see where leads are within the pipeline - easy, at first glance, to understand what's in the queue for your team. And it's flexible for your team to work together on pushing these deals forward.



Build Centralized Database using notion

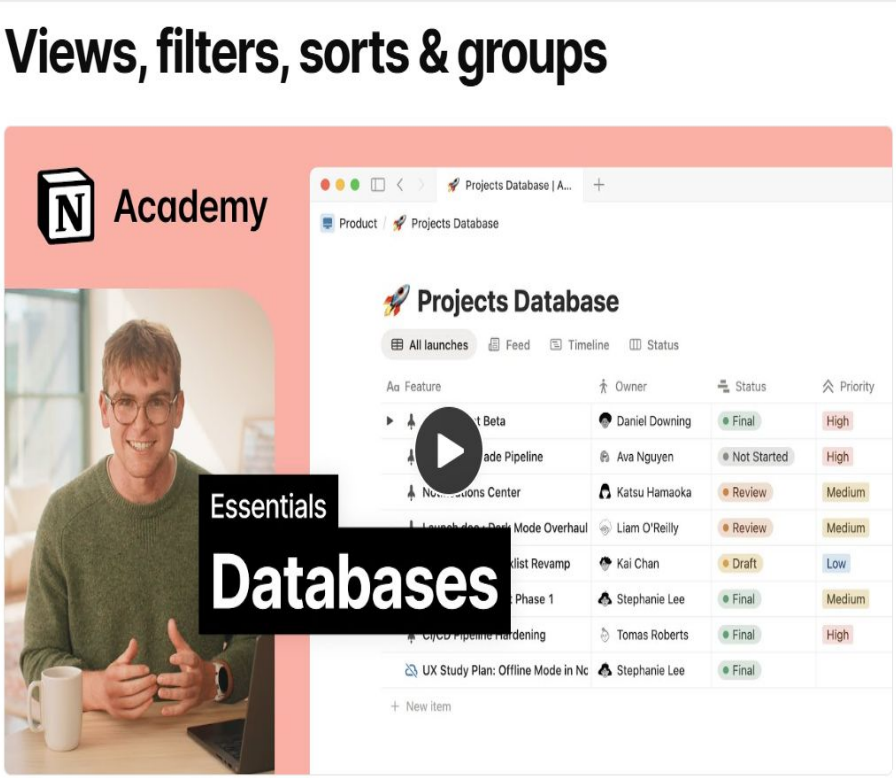
More details can be founded below:

Creating database:
https://www.notion.com/help/guides/creating-a-database

Views, filters, sorts & groups:
https://www.notion.com/help/views-filters-and-sorts

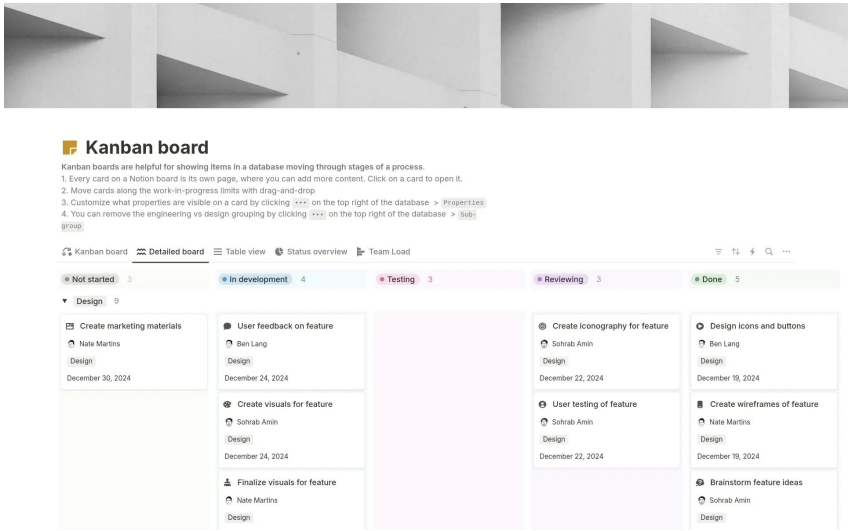
Import csv file:
https://www.notion.com/help/import-data-into-notion

Import, Merge, and Export CSV Files:
https://theorganizednotebook.com/blogs/blog/how-to-mport-merge-export-csv-files-notion?srsltid=AfmBOoq7JaWSwsdPzt7Kr5Uw5fAIYx8kKVla3un_5xlVXscst2r4t8PT



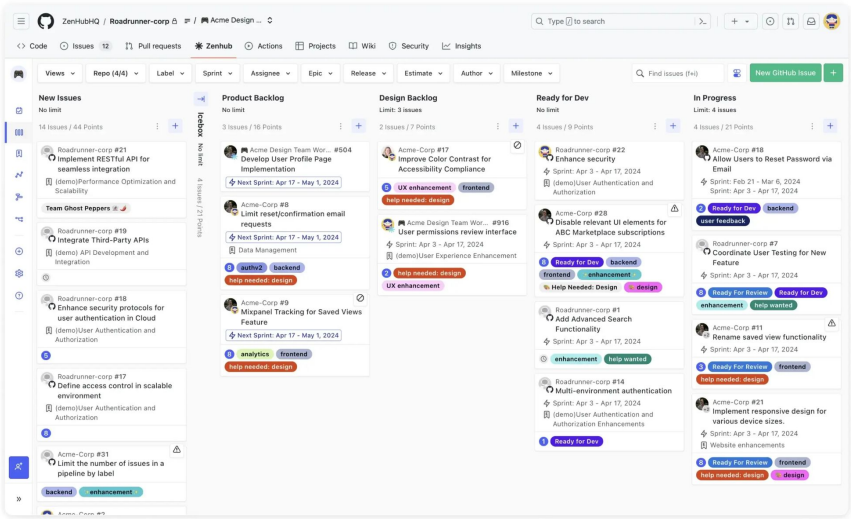
Project Management Tools

Notion Kanban Template



https://www.notion.com/templates/category/kanban?srsltid=AYCX9vIhdzxgGK9H6jtPrquc0ghXP3fNcWfZ7SvJywALgiFNIpKXcUCG

Github



https://docs.github.com/en/issues/planning-and-tracking-with-projects/learning-about-projects/quickstart-for-projects

https://docs.github.com/en/issues/planning-and-tracking-with-projects/customizing-views-in-your-project/changing-the-layout-of-a-view

Project Risk – Definition

"Project risk is an uncertain event or condition that, if it occurs, has a positive or negative effect on a project objective"

"A combination of the probability of a defined **threat** or **opportunity (Likelihood)** and the magnitude of the consequences of the occurrence (Impact) defines a Risk Index"

Tasks	Threat	Reason
Data Collection	Delay 2 days	Business Scope Always changing
Data Cleaning	Delay 2 days	Business Scope Always changing
Machine Learning Model	Delay 0 days	Nil

Risk Impact

Threat → Scope → **Poor Quality Product**
Threat → Schedule → **Late Delivery**
Threat → Cost → **Overspend**

In addition there are health, safety and environmental threats that **must** be managed

Identify Risks
Assess likelihood and impact
Rank risks and prioritize
Define risk management approach & actions
Implement actions
Monitor & review

Make the management of risk integral to the way the project is managed
Ensure that cost and time contingencies are consistent with identified risks
Focus on the “significant few” – don’t try to manage too many risks
Be vigilant (be ready for possible problems and diffuculties)and proactive

Project Monitoring and Control



Project Monitoring

Typical Monitoring Activities

- regular reviews of progress against schedule using WBS as basis (Plan against Baseline)
- regular review of actual costs (O/P from SAP) against budgeted costs and Earned Value at WBS level
- regular review of resource loading
- regular progress meetings with project team
- regular meetings with contractors
- production of periodic progress reports
- risk reviews
- inspections/ audits

Project monitoring checks whether that standard is actually being followed and whether the project is achieving its objectives. In short, SOPs are the “rulebook,” and monitoring is the “inspection and feedback loop.”

Project Control

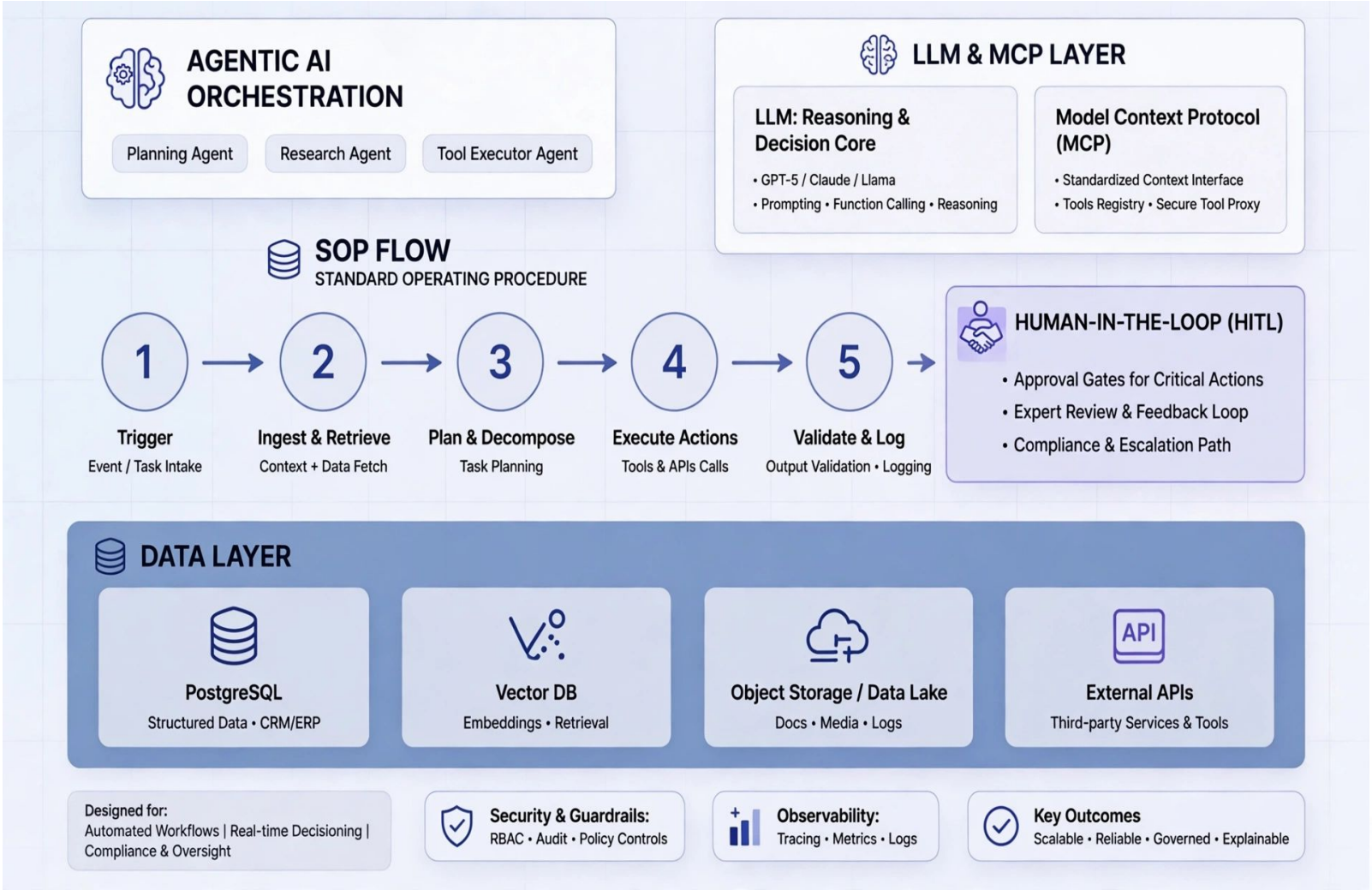
Typical Control Activities

- assign responsibilities at Work Package level
- staged authorisation of work to be done
- staged release of budgets (staged release of WBS(e) numbers)
- ensure PM has a ‘Management Reserve’ under his control
- seek corrective action reports when WPs go ‘off track’ (overrunning or overspending)
- release Management Reserve carefully

Project Monitoring and Control Summary

Monitor against the plan – status regularly
Take a factual approach to decisions
Identify management action early
Check that defined controls are being applied – correct if necessary
Apply change control

Full Agentic AI Workflow Automation

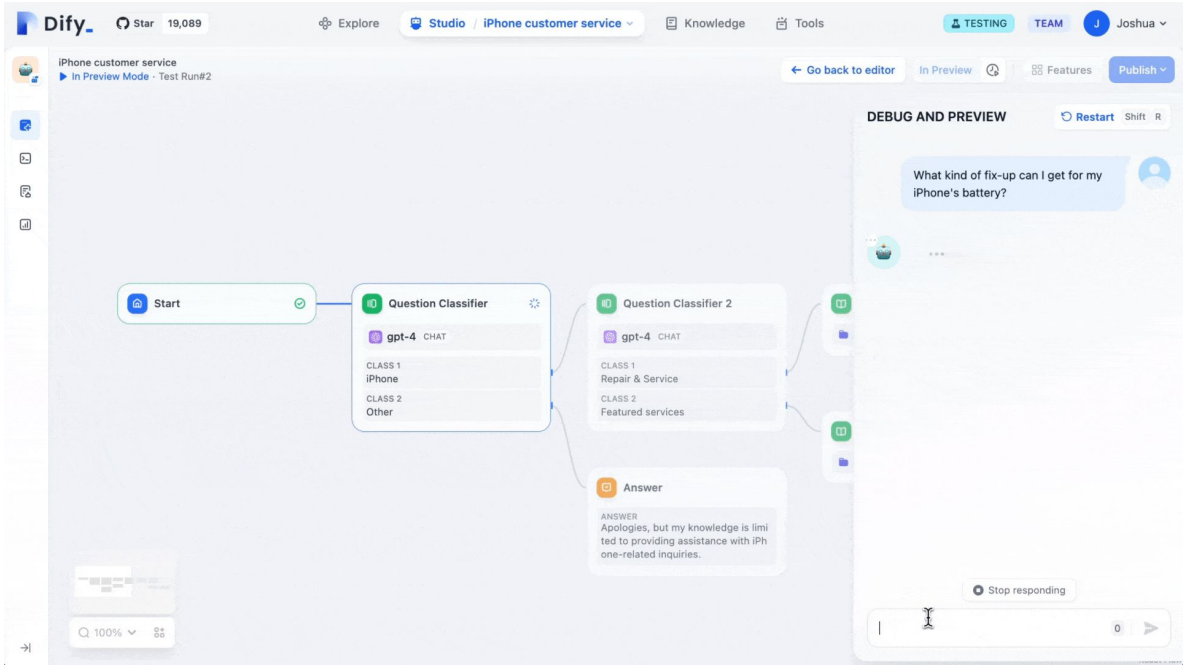


Agentic Workflow NoCode Tools 1: Dify

The Platform for Production-Ready Agentic Workflows.

Features:

Build Agentic workflows, RAG pipelines, with rich AI model and tool support on one collaborative workspace. Deploy on cloud, VPC, or self-hosted, so teams move from prototype to production without rebuilding the stack.



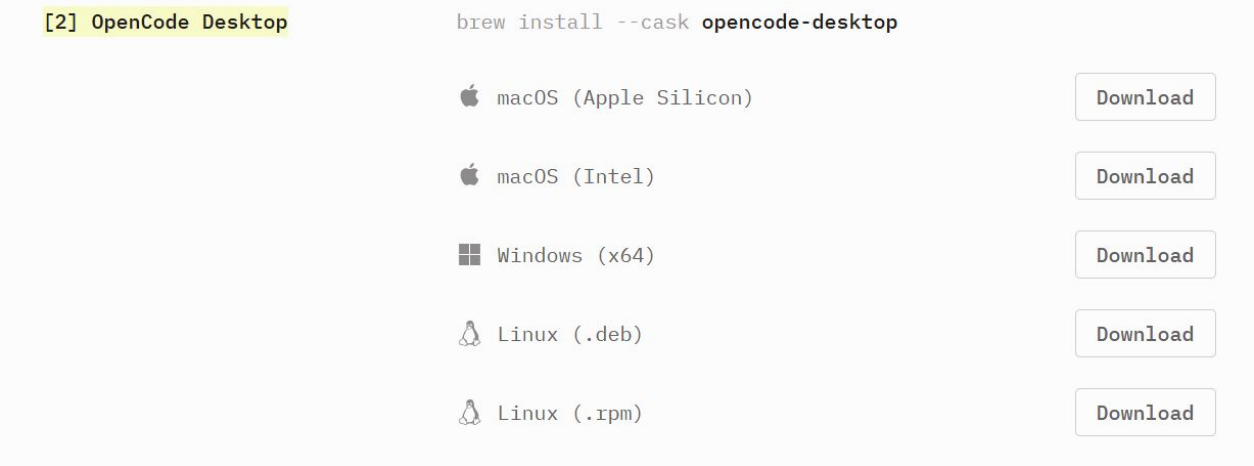
https://dify.ai/

OpenCode Installation

Make sure your notebooks have enough space for 2 GB for downloading both OpenCode and Ollama.

OpenCode Size is around 226MB

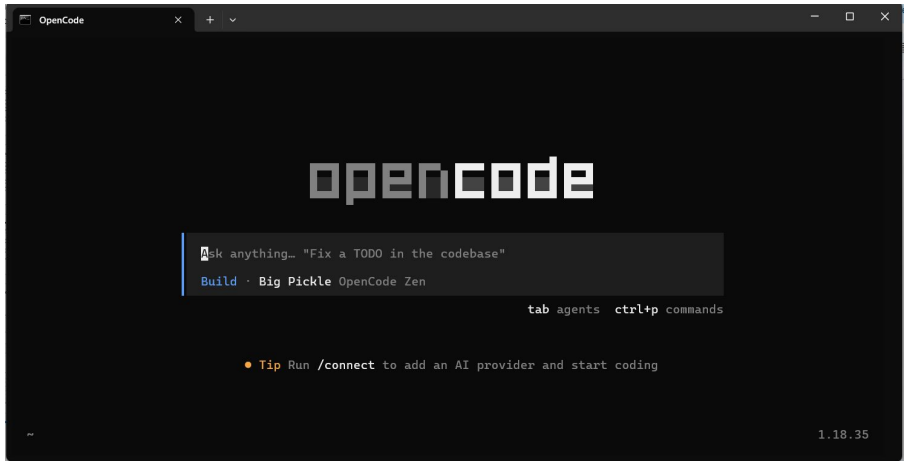
1. Open https://opencode.ai/download
2. In [2] OpenCode Desktop, Click “Download”

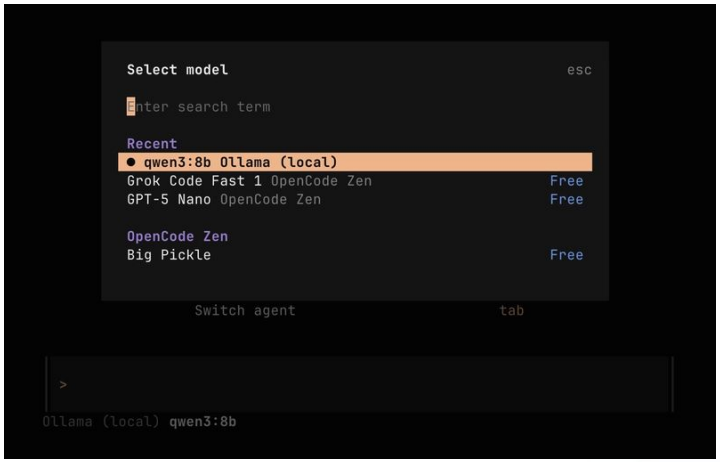


3. After Download, install OpenCode Desktop to local PC.

OpenCode LLM Provider Connector Installation

1. Open Opencode
2. Type “/connect”
3. Select OpenCode Zen, sign in, copy your API key, and paste it into the prompt to access free tier models .
4. Select Big Pickle as LLM model.





Introduction of Python Framework

Flask - Flask is a microframework for Python. The main purpose is to develop a strong web application base. As compared to Django, Flask is best suited for small and easy projects.

Django - Django is a high-level Python Web application development framework that encourages us to develop things rapidly. It's free and open source.

Web2Py - It is a free open source full-stack development framework in python which allows the user to develop things quickly. It is a cross-platform framework that supports all popular operating systems.



Commonly used Flash Framework



Comparison between Django & Flask

	django	Flask
Type of Framework	Full Stack Web Framework	WSGI framework
Flexibility	Feature-packed	Full flexibility
ORM Usage	Built-in ORM	SQLAlchemy is used
Design	Batteries-included	Minimalistic design
Working Style	Monolithic	Diversified

What is Dash / Python Flask Framework?

- **Dash** is **Python framework** for building web applications. It is open source, and its app run on the web browser

- It built on top of **Flask**, Plotly.js, React and React Js.

- It enables you to build dashboards using pure **Python**.

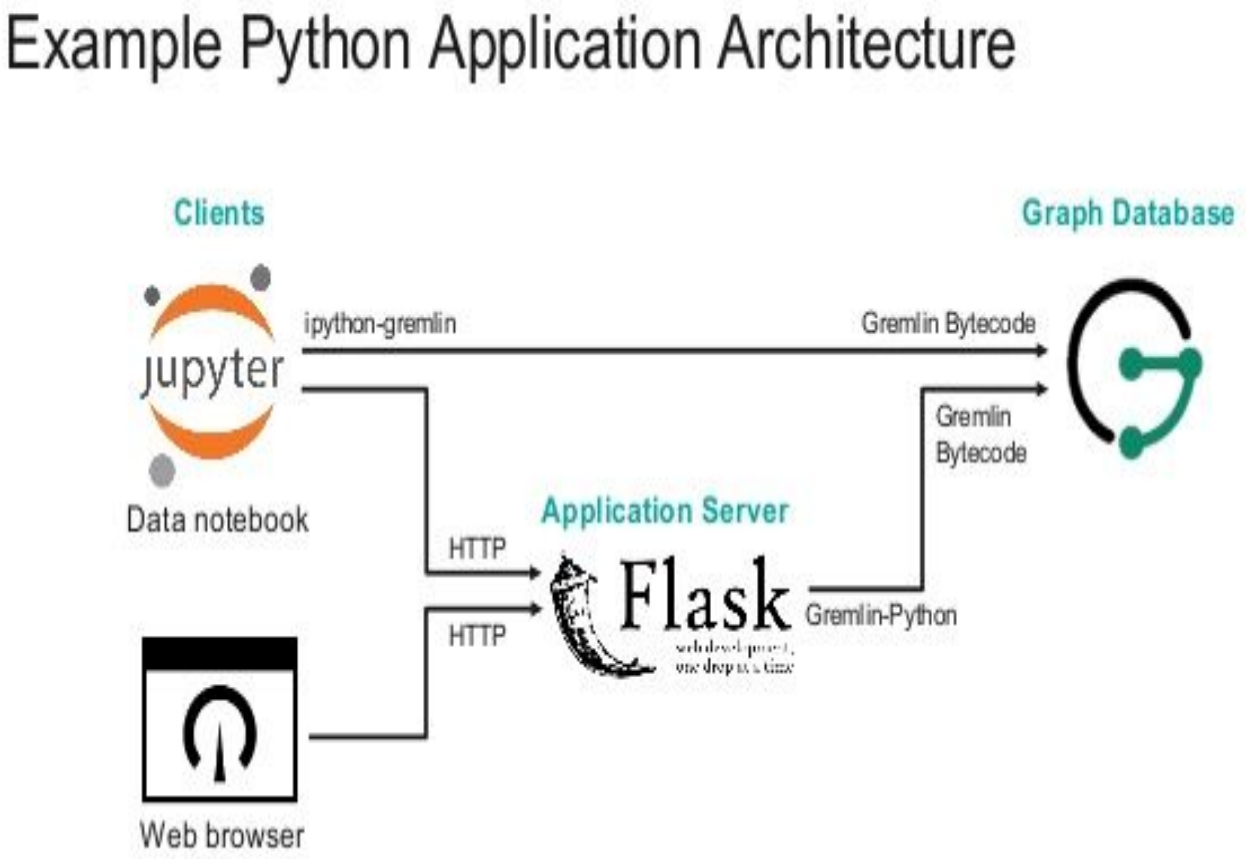
What is Dash / Python Flask Framework?

Every Dash App is composed of two main parts:

Layout = is used to define how all the content should be laid out on the application.

Callbacks = are used to make each of the desired components of the Dashboard interactive.

Python Flask Framework Architecture



Web Application Implementation using OpenCode

Objective

- Build an web application to demonstration KPIs using dashboard
-

Flow

1. Open Opencode
2. Enter following prompt

Build web application using python flask framework to conduct dashboard that include features a. customer login, b. chart and graphs with KPIs, c. detailed records with searching, filter, sorting and pagination and d. export records in CSV/PDF files.

3. Wait for 10-15 minutes that an new folder with your project is created.
4. Study the documentation in md format to run the web application.

HKUSPACE
香港大學專業進修學院
HKU School of Professional and Continuing Education

THANK YOU

